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================================================================================
   SCARLET REEL
   STUDIO MODE  —  INSTRUCTION MANUAL  (DETAILED)
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   Build, direct, and publish your own pulp-noir cases.

   UPDATED 10 SEPTEMBER 2026 — HD walking-room / portable-media pass
   ----------------------------------------------------------------
   Use the Stage node (SceneNode.stage) for new point-and-click rooms. The
   older scene.stage2d format mentioned below is imported into a Stage node.
   New in this pass: deliberate-click trigger policy, spawn/exit and pickup
   approach audits, gated actors and optional markers, feet-anchored scaling,
   scroll offsets and a properly masked spinner cockpit. See Sections 19–21.
   Large exports show a bounded PREVIEW; Download / Copy contain the full
   file. Never try to import the truncated preview. HD library reads are
   cursor-based so one enormous IndexedDB response cannot crash the browser.
   Browser tests, not just graph validation, are required before shipping.

   This manual is written from the actual Studio source (src/studio/Studio.tsx,
   src/game/Game.tsx, src/logic.ts, src/types.ts), so every term, field and
   behaviour below is what the tool actually does — not a paraphrase.

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   RECENT CHANGES (press the lie, walkable 2D scenes, craft linter,
                   flag notes, route charts, HUD, present-item verb)
   — read this first
--------------------------------------------------------------------------------
   NEW: PRESS THE LIE □        the contradiction verb. A dialogue beat can
   (catch them in a lie)         carry a LIE: while that line is on screen the
                                 player gets a "□ Press them on that" button,
                                 opens their satchel, and must present the ONE
                                 clue that proves the claim false. Right clue
                                 -> the CAUGHT beat plays (auto-gated on the
                                 caught flag), trust/meters swing, optional
                                 "…will remember that" card. Wrong clue -> a
                                 BLUFF-CALLED line, heat up / trust down, and
                                 after a wrong-evidence LIMIT (default 2) the
                                 claim shuts for the scene. In an INTERROGATION
                                 scene set "composure damage" and a proven lie
                                 shatters the suspect (crack/clamp by trust).
                                 Authored on the beat via the Dialogue tab's
                                 "□ Press the Lie" panel. The validator checks
                                 the evidence exists/obtains and the caught
                                 beat is gated. See Section 11 "PRESS THE LIE".
                                 Try Demo Lab: Press the Lie.
   NEW: WALKABLE STAGE □       choose the Stage node type. Ground clicks or
                                 arrow/WASD keys move the character. In Stage
                                 -> Interaction choose hotspotTrigger:
                                 aim = explicit clicks only (recommended for
                                 adventure games); smart = exits need clicks,
                                 ordinary objects can fire on walk-over;
                                 walkover = opt-in legacy automatic triggers.
                                 A chosen target owns its walk-up. Steering
                                 or clicking empty floor cancels that target.
                                 State, gates and consequences remain shared
                                 with the graphic-novel engine. Section 19.
   NEW: CRAFT LINTER            the Continuity Department now also flags STORY
                                 CRAFT, not just broken graphs: flags that are
                                 set but never read (a payoff that never lands),
                                 "will remember that" cards with nothing behind
                                 them, memory tags nobody recalls, quick-time
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	deadlines that change nothing when they expire, two choices that answer identically, and long passive corridors with no interaction. All are advisory (gold warnings); they never block publish. See Section 14.
NEW: FLAGS TAB (□)	a story-level tab listing every flag used anywhere (choices, gating, hotspots, QTE timeouts, pattern puzzles, the corkboard, on-enter effects), each with a NOTE box and a "set in …" line. Notes double as content — see FLAG-KEYED RECALLS below.
NEW: FLAG-KEYED RECALLS	a beat's "Memories recalled" key and the gating "Remembers" picker now accept FLAG NAMES as well as memoryLog tags. When a flag (not a memory) matches, {said} echoes the flag's NOTE from the Flags tab — "she remembers what you SAW", not just what you said. The validator accepts both.
NEW: PRESENT-ITEM VERB	a choice can PRESENT an item to the character ("show the photograph"). It locks (or stays hidden) until you hold the item, optionally HANDS IT OVER (leaves the satchel), and records an `item:<id>` memory so later beats/gates can recall or remember the presentation — {said} = the item's name. See Section 10 "PRESENT ITEM".
NEW: INTERROGATION	a scene can run a COMPOSURE-PRESSURE interrogation: attitude-tagged replies press the suspect (thunk + gauge jolt + reaction), composure drops, and at zero their TRUST decides — CRACK (★ stamp + broke beat + flag + swing) or CLAMP (✖ stamp + backfire). URGENT: hesitate and the suspect STEADIES (a visible creep + one +1), a live "break them now → talk / shut down" prediction, a red vignette + heartbeat at critical. See Section 10 "INTERROGATION".
NEW: EXAMINE (hold to inspect)	a giveItem/showText hotspot can be a HOLD-TO-INSPECT zone: the camera zooms in, an "EXAMINING" meter fills over the authored hold, then the pickup/description fires and the zone is hidden. Pair the action's flag with a gated beat to narrate the find. See Section 9 "EXAMINE" + The Neon Diner's "Mabel's Station".
NEW: ROUTE NOTES	the end-of-part Flowchart and the end-of-act RouteMap now list the flags the stretch just played set, with their notes.
NEW: CASE PROGRESS	the in-game Case Progress panel gained a "Story so far — flags set" list (note, or the flag's readable name).
NEW: PART BREAKS IN STUDIO	the Scene tab has an "End-of-part route chart" field — part breaks no longer require hand-editing story data. Use a choke point the player must pass through.
CHANGED: BOTTOM HUD	the "— The Scene —" label and "□ narrate" button are gone; dialogue now sits in a cinema frame (shimmering hairline, corner ticks) with a per-speaker accent nameplate, a pulsing continue pill, and sheen-swept choice buttons. All motion honours reduced-motion.
FIXED: HEAT BUST	the bust now fires the MOMENT Heat reaches

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                              the limit — including from reply choices,
                              pattern-puzzle outcomes and on-enter effects,
                              which used to wait for the next scene change.
   FIXED: QTE UNDER THE MENU  quick-time clocks pause while the ☰ Menu is
                              open and resume from the paused value.
   FIXED: INTERSTITIAL KEYS   the part-break chart closes on Esc/Enter/
                              Space like its siblings, and the global
                              shortcuts stand down while a chart / route
                              map / act card is up (no more firing the
                              old scene's choices behind the overlay).
   FIXED: VALIDATOR           a bad hotspot gate was reported twice; it's
                              now once. Puzzle rewards (and QTE-timeout
                              items) count as grantable, so a gate on a
                              puzzle-granted item no longer false-alarms.
   SAVED: SAVE SCHEMA         save slots now carry a schema stamp; old
                              un-stamped saves load and get migrated, and
                              saves from a newer build are refused rather
                              than mis-loaded.

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================================================================================
   SECTION 1  —  WHAT STUDIO IS & HOW IT WORKS
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   Studio is the built-in authoring tool for Scarlet Reel. It is a browser
   editor, not a separate app: the case you design lives as one JavaScript
   object (a "Story") that the game loop renders directly. There is no
   export-to-code step, no compile. What you set in Studio IS the game.

   THE PIPELINE
   ------------
   1. You build a Story in Studio.
   2. Studio autosaves the draft to browser storage (localStorage) every ~0.8s.
   3. You Validate (audit) the story, then Play draft to test it.
   4. When clean, you Publish — the newsstand now serves that exact version.
   5. You can Export a .reel.json to back it up or hand it to someone else.

   PLAYING EXACTLY WHAT YOU BUILT
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When a case is opened from the newsstand, the game loads in this order of priority:
1. the PUBLISHED copy (last time you clicked Publish),
2. else the live AUTOSAVED DRAFT,
3. else the BUILT-IN case bundled with the app.
So the moment you Publish, the newsstand reflects your latest Studio build.

ONE STORY = ONE OBJECT

A Story contains: scenes (nodes), characters (suspects/cast), items (clues & props), an optional deduction corkboard, an optional win/lose briefing, a layout map (scene positions on the canvas), and an optional array of embedded media assets for export/import. Everything else hangs off these.

NO DESTRUCTIVE SAVES

Studio writes drafts under a per-story key and keeps a history stack for Undo/Redo. The built-in case is never overwritten until you explicitly Publish, and Reset restores the built-in version of the current story.

SECTION 2 — GETTING STARTED

HOW TO OPEN STUDIO

- On the Newsstand (main menu), click the "Studio Mode — build your own reel" button in the bottom bar.
- Studio opens on the last story you were editing. (Its draft is loaded from browser storage; if none exists it loads the built-in version of the first story.)

THE LAYOUT

TOOLBAR (top bar)
Story selector, title, Undo/Redo, node tools (Scene / Credits / Cutscene / End), New Story, Check (validator), Repair, Export, Import, Share, Reset, Storage, Delete, Play, Exit. Grouped with thin dividers into logical clusters (story / history / add nodes / check / file / danger / primary actions). Uses a clean sans-serif font (not the pulp slab).

CANVAS (centre)
The story graph. Scene cards are nodes; the handles on their edges are the exits you drag to other cards. Right-click for context menus. This is the "flowchart" of your case.

INSPECTOR (right panel)
The editing surface for whatever node is selected. Fifteen tabs:
Scene, Beats, Hotspots, Choices, Test, People, Audio, Items, Win/Lose, Acts, Episodes, Media, AI Images, Dialogue Box, Theme.
Above the tabs sit three collapsible summary panels:
Creator workflow (the 5-step checklist, Section 17)
Ship Readiness (Start scene / Ending scene / No hard errors / Case objective)
Quick Reference (story + character + item counts, and which scene you are editing)
The inspector width is draggable and persisted.

EVERYTHING STARTS MINIMISED
Every collapsible panel (the summary panels, the cinematic Camera / Foreground overlay / Lighting, and each dialogue beat, choice and hotspot card in the Dialogue / Choices / Hotspots tabs) opens MINIMISED by default — just a title bar with a ▸ to expand it. Click a title bar to maximise a section, click it again to minimise. Your open/closed choices are saved

per panel so the inspector stays the way you like it between sessions. A freshly-added beat or choice (or a choice routed in from the canvas) automatically opens so you can edit it immediately; the rest stay tucked away.

AUTOSAVE

- The draft is saved ~0.8s after any change. A "autosave on / autosaved" indicator appears in the toolbar.
- If browser storage is full, Studio shows a warning and offers the Storage button to reclaim stale data; the draft is retried once.
- Media is stored as compact "asset:<id>" references, never raw base64, so large audio/video uploads don't blow the quota.

SECTION 3 — THE TOOLBAR — EVERY BUTTON, IN DEPTH

THEME

Opens the global title page editor (not the current story). Controls the main menu itself:

- title text (e.g. "Scarlet Reel")
- tagline (e.g. "A Dime-Store Detective Serial")
- main soundtrack (paste a URL or upload an MP3)
- title background image
- neon glow colour (a HEX value for the flicker)

Changes apply instantly to the newsstand title screen.

STORY SELECTOR (dropdown)

Lists every case. Picking one loads its draft (or the built-in version if there's no draft). Resets the Undo/Redo history for the new story.

TITLE INPUT

Renames the current case. Persists to the draft on the next autosave.

UNDO / REDO

Step through the edit history. History is held per session, capped at the last ~100 states. Undo/Redo also work with Ctrl+Z and Ctrl+Shift+Z / Ctrl+Y.

+ NEW STORY

Creates a new, empty case with a pre-filled Case File (objective, empty win/lose conditions). It has no scenes yet. You're dropped on the briefing tab first, then add your first scene. A blank case does NOT throw a wall of validator errors — it's treated as a guided draft, and Publish is blocked until a first scene exists and is marked as the start scene.

+ SCENE

Adds a standard game scene node at a random canvas position. Its title, background, prose, choices, hotspots and dialogue start empty.

CREDITS

Adds an opening/ending cinema-credits node. It hides the main HUD and renders a film-style intro (big title, subtitle, divider, credit lines) that can fade-up, type-writer, or SLAM in (oversized, snapping to size — the pulp title-card punch), and either stagger, roll, or play as one-at-a-time CARDS over a MONTAGE of stills. The montage is now a visual stills editor: add images via the image browser (browse all art + uploads with hover preview) or upload, reorder with ▲/▼, remove with ×. The montage can cross-dissolve or hard-CUT between stills (the Tarantino title-sequence bang — add black spacer frames as "gradient:…" entries for the chapter-card rhythm), and the title itself can take a custom TITLE COLOUR (the glow re-tints to match; stark yellow over hard cuts is the

classic look). Defaults fall back to the scene's prose line-by-line. The "type" font option is now Newsreader (clean literary serif).

CUTSCENE
Adds a full-screen video/storyboard cutscene node. Hides the main HUD and plays a video background. Used for cinematic beats (e.g. a talking-head clip) with a centred Continue button.

END
Adds an ending node. An ending takes no output handle. You set its tier (victory-gold / victory-silver / gameover), headline and recap, which are shown on the final reel panel and recorded to the player's ending gallery.

CHECK (VALIDATOR)
Toggles the "Continuity Department" panel at the bottom of the canvas. It lists every issue, warning and error, with a red/gold severity. Click an issue to select and fly the camera to that scene.

REPAIR
One-click auto-repair of the most common authoring faults. See Section 14 for the exact list of what it fixes.

EXPORT
Bundles the current case (plus every uploaded image, voice line and video) into a self-contained .reel.json. Opens a dialog to Download or Copy.

IMPORT
Loads a .reel.json. Restores scenes, choices, characters, items and every embedded media asset, then saves the compacted story as your draft.

SHARE
Builds a link that encodes the whole story into the URL hash (#reel=...). Copies it to the clipboard. Best for small/medium cases; very large cases are better exported as .reel.json.

PUBLISH
Compacts the draft and pushes it to the newsstand. If it's a blank case (no scenes) it blocks; if there are hard errors it blocks and opens the Validator. If storage is full it reclaims stale backups and retries once.

RESET
Discards the current draft and restores the built-in version of the selected story. Asks for confirmation first.

STORAGE
Scans browser storage for stale "-backup-pre-r…" migration keys and removes them, reporting how many KB it freed. Useful when autosave hits quota.

DELETE
Removes the current story, its draft and its published copy. It first asks whether to also delete the story's media assets from the library, or keep them for reuse elsewhere. At least one story must always remain.

PLAY
Launches the current draft in the game loop. Blocked while any hard error exists (opens the Validator). This is the fastest end-to-end test of the story graph, gating and endings.

? HELP
Opens the field manual inside Studio: the pipeline, the five-step workflow, the gating/consequences contract, choices & quick-time, dialogue beats & memory recalls, hotspots, meters/trust/perks, the Validator, keyboard shortcuts, the troubleshooting cheat sheet and the

softlock checklist. "Restore all help notes" brings back every inline ? note you dismissed; "Full manual" downloads the complete text. Small gold "? Help" notes also sit on each inspector tab (Scene, Dialogue, Hotspots, Choices, Test, People, Items, Media, Audio, Win/Lose) — each one is dismissable and remembered per note.

EXIT
Leaves Studio back to the newsstand.

SECTION 4 — THE STORY GRAPH (CANVAS)

WHAT IT IS

A node graph of scene cards. Each card is one SceneNode. Exits are the outbound handles on the right edge of a card; you drag one onto another card to create an edge (a connection). An edge is either a choice or a hotspot that moves the player.

CARD ANATOMY

Header:
- a badge: START (*), END (flag), or CREDITS / CUTSCENE for those node types.
- the scene title.
- an issue count (a red warning glyph and a number) — how many validator issues this scene has. Hover to read.
- Delete (x) — removes the node (not allowed for the start scene).

Body preview:
- the background image/gradient (or a video icon), tagged "NO INPUT" if it's the start scene and "NO OUTPUT" if it's an ending.

Footer:
- "N hotspots · M beats" count.
- one row per outgoing choice: an attitude-coloured arrow + the choice text.
- one row per outgoing hotspot (a gold target icon + label).
- "drag to add choice ->" — a self-terminating universal handle.

NODE TYPES

STANDARD	The normal playable scene: full HUD (meters, satchel, case file), hotspots, choices, dialogue.
CREDITS	Film-style title card, no HUD. Big type + optional rolling credits.
CUTSCENE	Full-screen video/storyboard, no HUD, centred Continue button.
ENDING	Terminal. Has endingData (tier/headline/recap) and no output handle.

CONNECTING THE PLOT

- Each outgoing handle maps to a specific choice or hotspot. Choice handles are coloured by attitude; hotspot handles are gold.
- Drag handle -> drop onto another card to set that choice/hotspot's target scene.
- Dropping onto the "drag to add choice ->" handle creates a brand-new choice that targets that scene and opens the Choices tab with it focused (so you can immediately write its text).
- Clicking an existing edge selects its source scene and opens the correct editor tab (Choices or Hotspots), so you can edit the wiring instantly.

RETURN KEY BEHAVIOURS

- An edge whose target scene no longer exists becomes "dangling"; the validator flags it. Repairs re-point it to the start scene.
- A scene that is NOT an ending and has NO exits is a dead end. The validator

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    flags it as an error. Repair adds a guaranteed open exit.
  - A scene whose every exit is gated is flagged as a warning ("confirm the
    required item/flag can be obtained first").

  RIGHT-CLICK MENUS
  -----------------
  On a scene card:
    - Edit scene            (selects the node and opens the Scene tab)
    - Duplicate node        (copies the scene with fresh node/choice/hotspot/beat ids)
    - Copy node             (places it on the Studio clipboard; paste the copy elsewhere)
    - Paste next to it      (pastes the clipboard copy beside this card)
    - Set as start scene / This is the start scene
    - Mark as ending / Remove ending flag
    - Delete node
  On empty canvas (right-click):
    - Add scene here / Add credits here / Add cutscene here / Add end scene here
    - Paste "<copied title>" here
    - Fit view

  TELEPORTING
  -----------
  Drag a card to reposition it (the layout is saved). Zoom with the mouse wheel,
  pan by dragging empty space, and use Fit view to frame the whole graph. The
  minimap (bottom-right) gives an overview; node colours show endings vs. normal.


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  SECTION 5  —  THE DATA MODEL (THE BUILDING BLOCKS OF A CASE)
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  A case is one Story object. Knowing the fields is the fastest way to author
  well, because the inspector is literally editing these.

  STORY
  -----
  id                  unique key (used for saves, endings and storage keys)
  title               case name (newsstand card)
  tagline             one-line hook
  coverImage          newsstand cover (image or gradient:)
  startSceneId        which scene the case begins in
  scenes[]            all SceneNodes
  items[]             all Items
  characters[]        all Characters
  deduction?          an optional corkboard (suspectIds + connections)
  briefing?           an optional Case File (objective, win/lose, heat)
  layout?             map of sceneId -> {x,y} canvas positions
  assets?             transport-only; embedded media during export/import

  SCENENODE
  ---------
  id, title
  nodeType            "standard" | "credits" | "cutscene"
  background          url | gradient: | video:<url> (or uploaded asset ref)
  ambientAudio        "none"|"rain"|"jazz"|"drone"|"tension" (synth loop)
  particles           "none"|"rain"|"smoke"|"storm" (auto from audio if unset)
  musicUrl / sceneAudioUrl   looping background music file
  musicMode           "inherit"|"track"|"none"
  narrationAudioUrl   a recorded MP3 that replaces Web Speech for the prose
  autoNarrate         default true; false = never auto-narrate
  noTypewriter        default false; true = show prose instantly (for MP3 sync)
  choiceDelaySeconds  a hold timer before choices/continue appear (for cinematics)
  credits?            credits settings (title, subtitle, lines, entrances, style)
  prose               the narrator's opening text
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dialogueBeats[]     the spoken/scene lines
hotspots[]          clickable regions
choices[]           menu choices / navigation / replies
onEnterActions[]    consequences fired once on first entry
isEnding            true = terminal ending node
endingData          { tier, headline, recap }
cinematic?          camera/lighting/foreground settings for the background

ITEM
----
id, name, icon (emoji glyph), iconUrl (optional image), description,
flavorText, isClue (true = appears on the corkboard), combinableWith?
  combinableWith = { targetItemId, resultItemId }  -> a crafting recipe.

CHARACTER
---------
id, name, role, avatarUrl (image or emoji:), trustMeter (starting trust),
voice?  { voiceName?, pitch?, rate? }  (Web Speech settings).

CHOICE
------
id, text, targetSceneId
attitudeTag        "Neutral"|"Sultry"|"Aggressive"|"Clever"
gating?            rules that must pass to be pickable
consequences?      what it does when picked
reply?             true = answer in place, never navigates
hidden?            true = invisible until its gate passes
timeLimitSeconds?  a countdown; the choice expires at 0
isRepeatable?      default derived (see Section 10)
exclusiveGroup?    choosing one resolves all choices sharing this group

HOTSPOT
-------
id, label, cursor ("look"|"take"|"talk"|"use"|"go"),
bounds {x,y,w,h} (percent of the scene viewport),
isRepeatable, gating?, action, customFailMessage?, showItemImage?

HOTSPOT ACTION
--------------
type: "goToScene" | "giveItem" | "setFlag" | "showText" | "meterChange"
    | "openBoard" | "useItem" | "patternPuzzle"
targetSceneId?, itemGranted?, flagSet?, text?, meterChange?, itemConsumed?
patternPuzzle?   a PatternPuzzleConfig (the ledger mini-game — Section 9)

DIALOGUEBEAT
------------
id, characterId (or "narrator"), text, expression?,
conditionRule? (Gating — the beat only appears when this passes),
background? (a per-beat replacement image),
cinematic? (camera/lighting for that replacement image),
backgroundTransition ("cut"|"fade"|"crossfade"|"fade-black"|"slide-left"|"slide-right"|"iris-in"),
backgroundTransitionSpeed? (seconds),
audioUrl? (per-line MP3 replacing TTS), silent? (no narration at all).

GATING (the lock rules)  — used on choices, hotspots, dialogue beats
---------------------
requiresItems[]     must be holding all (or, with ALL off, at least one)
requiresFlags[]     must have all these flags active (or any)
excludesItems[]     must NOT be holding any of these
excludesFlags[]     must NOT have any of these flags
requiresAll         default true. false = ANY of the requires* satisfies the gate
minMeters           {Sultry?/Aggressive?/Clever?/Suspicion?} lower bounds
maxMeters           {..} upper bounds
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   minTrust            {characterId: number} per-character trust lower bound

   CONSEQUENCES (the effects) — used on choices, onEnterActions
   ------------
   setFlags[]       activate these flags
   clearFlags[]     deactivate these flags
   addItems[]       grant these items (once)
   removeItems[]    remove these items from inventory
   meterChanges[]   [{meter, delta}]  (clamped 0..10)
   trustChanges[]   [{characterId, delta}] (clamped -5..5)

   ENDINGDATA
   ----------
   tier     "victory-gold" | "victory-silver" | "gameover"
   headline the pulp headline shown on the final reel
   recap    a short what-happened summary

   DEDUCTIONBOARD (corkboard)
   -------------------------
   suspectIds[]        which characters can appear as nodes
   connections[]       [{a, b, flagSet, revelation}]:
                         a and b are clue items or characters; one should be a
                         clue and one a suspect. When you string a to b it sets
                         flagSet and reveals the revelation. (The engine only
                         accepts clue-to-suspect endpoints.)
   When every connection is solved, the engine sets the implicit "case_cracked"
   flag and grants +1 Deduce.

   CASEBRIEFING (Case File)
   ------------------------
   objective          one-line goal (players see it in-game)
   winConditions[]    how to win (list shown in the Case File panel)
   loseConditions[]   how you lose
   heatLimit          Heat value that triggers a bust; 0/undefined = off
   heatFailSceneId    the ending the player is sent to when heat maxes


================================================================================
   SECTION 6  —  INSPECTOR TABS, IN DEPTH
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   6.1  SCENE TAB
   --------------
   SCENE ID           read-only unique id.
   NODE LAYOUT TYPE   Standard game scene | Opening Credits | Cinematic Cutscene.
                      Credits/Cutscene switch the "hold" option too — a timer that
                      locks the Continue button and choices while the intro plays.
   TITLE              the name shown in the HUD.
   BACKGROUND         an image browser (searchable dropdown over every built-in
                      scene image in /img and /assets, plus all your uploads,
                      plus gradient presets — hover any name for a full preview).
                      Also accepts a path, "gradient:..." string, uploaded image
                      ("asset:<id>"), or "video:<url>" that overrides the image.
                      All image fields across Studio now use this browser.
   AMBIENT LOOP       none / rain / jazz / drone / tension (a synth atmosphere).
   PARTICLES          Auto (derived from the audio), none, rain, smoke, storm.
   PROSE              the narrator's opening paragraph (typewritten in-game).
   START SCENE        toggle — marks this node as the entry point.
   END-OF-PART ROUTE CHART   text field (e.g. "Part Two — The Long Drop").
                      When the player LEAVES this scene the game shows the
                      route flowchart of the stretch just played — and, if
                      the exit crosses an act boundary, the act's RouteMap
                      follows before the next act's title card. Leave empty
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                  for no part break. Use a CHOKE POINT the player must
                  pass through: a hub you backtrack through re-charts on
                  every exit (an exit with no new decisions charts
                  nothing and doesn't interrupt).
THIS SCENE IS AN ENDING   toggle — unlocks tier/headline/recap editing.
CINEMATIC CONTROLS  the full camera/lighting/foreground editor (Section 13).

6.2  DIALOGUE TAB (BEATS)
-------------------------
BEAT LIST          one collapsible block per beat (each title bar shows the
                   beat number, speaker and a snippet of the line — click it
                   to expand). Fields:
  SPEAKER          a character, or the reserved "Narrator (V.O.)".
  EXPRESSION       a free-text emotional cue, shown as a tag on the portrait.
  REPLACEMENT BACKGROUND  an image for just this beat, with its own:
                            TRANSITION  cut / fade / crossfade / fade-black /
                                         slide-left / slide-right / iris-in
                            SPEED       seconds for the transition (0.5 default)
  VOICE MP3 URL    per-line recorded audio; replaces TTS for that line only.
  SILENT           no narration at all (text only).
  LINE             the spoken/written text.
  CONDITION RULE   a Gating that controls whether this beat plays at all.
Add a beat, delete a beat (via the × in the beat's title bar), and re-order
freely. New beats open expanded so you can write them straight away.

6.3  HOTSPOTS TAB
-----------------
DRAW SURFACE       drag on the preview to draw a bounding box (percent). It
                   updates the selected hotspot, or creates a new one.
PER-HOTSPOT        one collapsible block per hotspot (its title bar doubles as
  CARD                the select / highlight — click to expand, click again to
                      collapse). Fields within:
  FIELDS             Label, Cursor, Action type, Target scene, Grants item,
                     Sets flag, Text shown, Consumes item, Repeatable,
                     Show item image, Custom fail message, Gating.

6.4  CHOICES TAB
----------------
CHOICE CARD        one collapsible block per choice (the title bar shows the
                   choice number, its attitude colour and a text snippet —
                   click to expand; the × deletes it). A choice routed in from
                   the canvas opens expanded automatically.
TEXT               the button label / menu text.
ATTITUDE           Neutral / Charm / Intimidate / Deduce (colours the button).
TARGET             where the choice navigates ("dangling" = no scene set).
REPLY              a conversational reply: does NOT navigate, just applies
                   consequences in place (Section 10).
HIDDEN             invisible until its gate passes (Section 10).
TIMER (s)          a countdown — the choice disappears when it hits 0.
REPEATABLE         one-shot vs. repeatable (default is derived; see Section 10).
EXCLUSIVE GROUP    picks that resolve one another (Section 10).
SET / CLEAR FLAGS  consequences on pick.
REWARD ITEM / CONSUME ITEM   inventory changes on pick.
METER DELTAS       per-meter changes (Charm/Intimidate/Deduce/Heat).
TRUST DELTAS       per-character trust movement.
GATING             the rules that must pass to be pickable (a collapsible
                   "▸ Gating rules" section you expand when needed).

6.5  TEST TAB (PLAYTESTER)
--------------------------
A simulation harness. You fake the world state with toggle chips and sliders:
  FLAGS        every flag referenced anywhere in the story.
  INVENTORY    every item in the story.
```

```
  METERS       Charm / Intimidate / Deduce / Heat (0..10).
  TRUST        per-character (-5..+5).
Then for the selected scene it runs a live check on every choice, hotspot and
dialogue beat, marking each available or blocked. Use it to confirm that a
gated path is actually reachable without replaying the whole case.

6.6  PEOPLE TAB (CHARACTERS)
----------------------------
Each character has: a portrait (upload, pick from the image browser, or an
emoji), a Name, a Role, a "Show this met character on corkboard" checkbox
(adds them as a suspect), and a Web Speech Voice block (pick a voice, tune
pitch & rate, and "Test voice" to audition it). Add and remove characters.

 TALKING ANIMATION (per character)
Each character also has a collapsible "Talking animation" section:
  TALKING FRAMES     sprite keyframes (transparent PNGs work best) that
                     auto-cycle as a portrait while this character speaks
                     in any scene. Add frames via the image browser (browse
                     all built-in art + uploads with hover preview) or
                     upload. Reorder with ▲/▼, remove with ×. A live
                     cycling preview shows the animation.
  RESTING POSE       a still image shown when this character is on-scene
                     but NOT the active speaker (falls back to the first
                     talking frame if unset).
  POSITION X / Y     horizontal position (% from left) and vertical
                     position (% up from the floor).
  SCALE              sprite size multiplier (default 1).
  FRAME TIME         how long each keyframe holds before swapping (seconds).
  MIRROR (FLIP)      mirror the sprite horizontally so two speakers can
                     face each other across the frame.

ON-STAGE BEHAVIOUR (in-game)
Characters who speak in a scene's dialogue beats automatically appear
on-stage (standing at their authored X/Y position, scaled and flipped as
set). The ACTIVE SPEAKER cycles through their talking keyframes; everyone
else holds their resting pose, slightly dimmed. When the dialogue ends
(choices appear), all characters hold their resting pose. Give characters
different X positions (and mirror one) so they can face each other.

PER-BEAT TOGGLE (Dialogue tab)
Each dialogue beat has a " Talking animation" checkbox (shown only when
the speaker has talking frames). Default ON; uncheck it and the speaker
shows their resting pose for that line instead of animating — handy for
quiet aside lines where mouth movement isn't wanted.

6.7  AUDIO TAB
--------------
SCENE MUSIC       inherit the previous track / play a looping track / no music.
TRACK URL & UPLOAD  set the looping MP3 (paste URL or upload to the library).
RECORDED PROSE NARRATION   a one-shot MP3 that replaces Web Speech for the
                           scene prose on first entry.
AUTO-NARRATE      toggle auto-play of the prose voice.

6.8  ITEMS TAB
--------------
Add items with an emoji icon (or uploaded image), name, description, flavor
text, and an "is clue" flag (shows on the corkboard). Recipes are set by
choosing a partner item and a result item — the two ingredients get combined
in the player's satchel by dragging one onto the other.

6.9  FLAGS TAB ()
-----------------
A story-level tab listing EVERY flag the engine can see — collected from
```

choices, gating (requires/excludes), hotspots, QTE timeout consequences, pattern-puzzle rewards and final fails, on-enter effects, and corkboard connections. Each flag gets:

FLAG ID	the machine name (snake_case).
READABLE NAME	the same, humanised ("saw_the_body" → "saw the body").
SET IN …	where the flag is set (scene — choice/hotspot/puzzle/ on-enter/board). Empty for a flag that is only REFERENCED in a gate — the validator will call that an impossible gate until you set it somewhere.
NOTE BOX	your authoring note for what the flag MEANS. Notes are player-visible content: a beat that recalls the flag echoes the note in place of {said}, the Route notes on the end-of-part / end-of-act charts show them, and the Case Progress "Story so far" lists each set flag with its note. Write them like sentences ("You saw the body in the study."). The validator warns about notes whose flag doesn't exist anywhere (typos, renamed flags).

6.10 WIN/LOSE (CASE FILE)

OBJECTIVE	one line (always shown in-game; players read it).
WIN CONDITIONS	bullet list shown in the Case File panel.
LOSE CONDITIONS	bullet list shown in the Case File panel.
HEAT LIMIT	Heat value that busts the case (0 = off).
BUST SCENE	the ending sent to when heat maxes (pick an ending node).
ENDING AUDIT	shows the count of victory-gold / victory-silver / gameover endings and flags any missing outcome.

6.11 MEDIA TOOLS (tab: □ MEDIA TOOLS)

Upload audio, video and images. Everything you upload is stored once per story (and is reusable in any scene/story via the image browser). A storage indicator tells you if uploads will survive a reload. Every image field in Studio now uses the IMAGE BROWSER — a searchable dropdown that lists every built-in image shipped in /img and /assets (all scenes, characters, covers, animation sprites) PLUS all your uploads PLUS gradient presets. Hover any name for a full-size rollover preview. Uploaded audio appears in the music/narration/voice pickers; uploaded video in the cutscene picker.

The tab also carries three image tools:

- □ BACKGROUND REMOVER — drop in ANY image (photo, scan, AI render, an existing sheet) and get a transparent PNG back. No grid to set, nothing to slice. EDGE FLOOD (default) is a magic wand: it floods inward from the border and removes only the backdrop CONNECTED to it, so a pale shirt survives a pale wall. COLOUR KEY removes every pixel near the picked colour — right for flat magenta or green-screen art. Click the preview to sample the backdrop under the cursor, tune tolerance/softness, tick "clean specks" and "trim to subject", then save to the library (□ cutouts) or download the PNG.

- □ SHEET SLICER — work out any sheet's grid and name its animations.

- ✂ SPRITE CUTOUT LAB — the same keying engine with full manual control, for building sprite assets from painted or photographic art.

6.12 AI IMAGES

A generative helper that produces noir-style art matching the game's look. A noir suffix is appended to every prompt to keep the aesthetic consistent. Generated images can be saved straight into the library or a given field.

6.13 THEME OPTIONS (GLOBAL TITLE PAGE)

```
----------------------------------------
Global (not per-story): title text, tagline, soundtrack, background image and
neon colour. Saved under a global settings key and applied instantly.
6.14  ADVENTURE MODE — POINT-AND-CLICK ROOMS
-------------------------------------------
The toolbar's kind switch (next to the title field) sets what KIND of game the
case is:  □ Noir Rail  (default — prose + choice rail, the house style) ·
□ Adventure  (walkable rooms, verb cursors, inventory puzzles) ·
□ Visual Novel (portraits and dialogue; reserved for now).

Picking ADVENTURE changes what NEW SCENES look like: every scene you add
comes out as a □ walkable STAGE node instead of a prose plate — keyboard on,
floor band, spawn point, proximity hints — and the status line tells you the
recipe: plate → □ obstacles → □ look / □ take / □ talk / ➔ go hotspots.

A ROOM IS:
  background           one painted room (the camera never moves)
  stage.floor          the strip the walker may stand on  (top..bottom %)
  stage.spawn          where the player enters
  stage.obstacles      rectangles {x,y,w,h} the walker ROUTES AROUND
  stage.playerSheet    the walker's sprite sheet (idle + walk animations)
  hotspots             the objects, each carrying a VERB CURSOR:
    look □ (showText) · take □ (giveItem) · talk □ (startDialogue) ·
    use ⚙ (useItem, consumed) · go ➔ (goToScene)
Hotspots can carry their own ART (hotspot.image, or hotspot.sheet + anim for
an animated object), so the bike you fix is drawn in the room instead of
described in prose. Cinematic foreground extras animate the room: the
Roadhouse sign flickers via anim: { preset: "flicker" }.

When the coin is ON the room is verb-first: only the marker you actually
click fires (brushing past the bar on the way out does not talk to the bar),
and a marker can answer several verbs — see each hotspot's "□ Verb coin —
other verbs" box. USE arms an item: pick ⚙ use, click the item in the rail,
then click what you want to use it on; objects list what they accept under
"□ Accepts these items (USE)", and anything else is refused with a line
(story-wide defaults: Flags tab → "□ Verb coin — refusal lines").

MULTI-STATE OBJECTS ("□ States", in each hotspot's inspector): ONE marker
that changes with the story — a lever up/down, a drawer full/empty, a door
locked/open — instead of stacking flag-gated duplicate hotspots. States are
checked top to bottom and the LAST one whose gate passes is the face the
player sees, so leave the unconditional one out (the object itself is the
default) or list it FIRST. A state overrides only what you fill in: label,
object image (or "no art in this state"), cursor, action (with its own voice
clip), and the lock (inherit the object's gate / always unlocked / its own
gate — the opened door needs no key). The zone, the verb answers and the
accepted items are shared, and the marker re-skins IN PLACE — it never
duplicates or resets the walk. The Check tab understands them: a state's
flag counts as produced, an unsatisfiable when-flag is an error, a state
route that leads to a scene is a real connection on the reel.

PERSPECTIVE: "Close-up size" / "Distance size" on the Walkable Stage panel set
how big the walker is at the front of the floor band vs the back (walking away
also covers less screen per second, so it doesn't read as skating).

WALK-BEHIND MASKS: on the Cinematic tab, tick "□ Draw this OVER the player
when they walk behind it" on any foreground image and drag its depth line.
While the walker's feet are above that line the cutout hides them — no second
mask asset, the cutout's own alpha does the job. Pair it with a barrier so
they can't walk through it either.

NO-GO BARRIERS: the Hotspots tab has a "□ Hotspots / □ Barriers (no-go)"
switch. In barrier mode drag boxes straight onto the plate — walls, counters,
```

a parked truck — and the walker routes around them; the plate shows the walkable floor band and every barrier. Click a barrier to select it, drag a new box to reshape it.

 DOORS: click a barrier in the list and give it a GATE. While the gate is shut the barrier is a solid wall (the walker physically cannot pass); the moment something opens the gate — a key hotspot, a conversation choice, a puzzle, a multi-state object's action — the barrier stops existing and the walker can go through. Lock-and-key made spatial; the Check tab errors on a door whose flag nothing can ever set.

Barriers are honest for EVERY kind of walk — clicks, floor taps, and the straight-line fallback all slide along blocked faces instead of ghosting through. Two details worth knowing as an author:
- A marker that sits BEHIND a shut gate (no route to it at all) refuses from wherever the player stands — its gate caption plays immediately, which is usually the clearest "come back later" signal you can give.
- Don't drop a spawn or entry-spawn point inside a barrier. The engine now rescues that (the walker steps out toward whichever side can reach the destination, and never snaps into a sealed sliver), but the rescue walk looks like a bug to a player. Spawn NEXT to the furniture.

PLAYER QoL KEYS (walkable rooms, automatic): H pulses every marker so the player can see what's interactive; DOUBLE-CLICKING a floor spot or marker jogs the walker there at 5x speed; 1-9 pick the verb coin's verbs in order; N walks to and talks with the nearest interactable character.

INTERACTABLE NPCS (Ambient NPCs, per actor): give an actor a "talk marker" link and the CHARACTER becomes that marker — their diamond hides, they get a chip, and clicking them (or pressing N) sends the walker after their LIVE position: the pursuit re-routes while they patrol and the conversation fires on catch. The Check tab errors on a link to a marker that doesn't exist and warns on one that isn't a conversation starter.

 WALK TEST (Stage tab, on the plate's toolbar): drops a LIVE walker onto the plate at spawn and drives it with the real engine router. Click the floor to see the planned route drawn dashed and walked; click a MARKER and the chip reports "✓ reached" — or "✗ No route" when your barriers/path seal it off, which is exactly the mistake it exists to catch. WASD/arrows drive it live, and the Doors toggle flips every gated barrier at once so you can prove the locked layout seals and the opened one flows. The test never touches story data.

 SENTENCE LINE (Interaction, per room): a SCUMM strip along the bottom reads the pending interaction as you hover — "Look at the drop-safe", "Use wrench on the bike", "Walk to" — and RIGHT-CLICK on an object is a quick look. Off by default; classic scenes never show it.

INVENTORY DISPLAY (Items tab → " Inventory display"): Classic (the chip rail every case already has — the default, so nothing changes), Container (a pouch / case / saddlebag sprite that pops open with the items laid out INSIDE it — supply your own transparent art, or use the shipped leather case), or Hidden (no dock). Pick the corner, columns, inset and heading.

DRAG AND DROP: drag an item onto a scene object to USE it there (in a walkable room the character walks over first, then uses it), or onto another item to combine the two. Works with a mouse or a finger. Objects answer with their own line when they don't accept what you dropped on them.

PLACING THE PLAYER: Hotspots tab → " Player start" — click or drag on the plate and the walker stands there (clamped to the walkable floor). The marker shows the real character sprite and stays ghosted while you place hotspots and barriers around them.

CURSOR ART (In-game theme tab → " Cursor art (per verb)"): give any verb its own cursor. A static image becomes the real mouse cursor (keep it under 128px); a SPRITE SHEET plays as a little animated sprite that follows the mouse (a reaching hand, a turning key). Set the hotspot point (pointing tip or centred reticle) and the size; "default" covers every verb you leave alone. No art at all = the game's system cursors, so nothing changes unless you opt in. The preview is drawn in your own in-game theme, at real size.

Inventory puzzles work as always — take things, combine them in the satchel (Item.combinableWith), then USE the result on a hotspot. Replies carrying presentItem: "..." are the "lay it on the bar" verb.

Worked example: RAIN SECTOR (an adventure case in the catalogue — verb coin, multi-state hotspots, walk-off-screen edge exits). Full write-up, including the things the mode still lacks: docs/adventure-mode.md

SECTION 7 — GATING RULES & CONSEQUENCES (THE ENGINE'S CONTRACT)

These are the two mechanisms that make a case dynamic. Gating decides WHAT IS AVAILABLE; Consequences decide WHAT CHANGES. Getting them right is the core of authoring.

In Studio every gate lives in a collapsible " Gating rules" section (on a choice, a hotspot, or a dialogue beat). It starts minimised so long choice/beat lists don't bury the main fields; the title bar shows whether the gate is "active" or "none set". Click it to expand and set the rules.

GATING — HOW A GATE IS EVALUATED

A gate returns true when every enabled clause passes. The clauses are:

requiresItems	player is holding ALL of these (or ANY, see ALL)
requiresFlags	ALL of these flags are active (or ANY, see ALL)
excludesItems	player is NOT holding any of these
excludesFlags	NONE of these flags are active
minMeters	each listed meter is AT LEAST its value
maxMeters	each listed meter is AT MOST its value
minTrust	each listed character's trust is AT LEAST its value

All clauses must pass. The "requiresAll" switch only affects the requires* clauses: with ALL on (default) you need every listed item AND every listed flag; with ALL OFF you only need ONE of them (item OR flag).

EXAMPLES

```
- require the item:      { requiresItems: ["cipher_note"] }
- require the item OR a flag:  { requiresItems: ["key"], requiresFlags: ["heisted"],
                         requiresAll: false }
- require high charm:    { minMeters: { Sultry: 4 } }
- require trust after a soft opener:  { minTrust: { lena: 1 } }
- lock out once hostile: { excludesFlags: ["harsh_start"] }
- require a deduction:   { requiresFlags: ["note_malveaux","ledger_malveaux"],
                         requiresAll: true }
```

CONSEQUENCES — WHAT HAPPENS ON PICK / ENTRY

setFlags	activate flags (used to unlock later gates and reveal beats)
clearFlags	deactivate flags (used to undo a harsher opening)
addItems	grant items (looked up from the story; duplicates are ignored)
removeItems	drop items from inventory

```
meterChanges   adjust meters; every value is CLAMPED to 0..10
trustChanges   adjust per-character trust; CLAMPED to -5..5
```

A consequence can be attached to a Choice and to an onEnterAction. Choices apply consequences on pick; onEnterActions apply ONLY on first entry to a scene (not on backtracking), so rewards and meters can't be farmed by looping in and out.

PROGRESSION FARMING

The engine deliberately keeps state-changing choices and hotspots one-shot by default (see Section 10). If you set a reward/meter choice to repeatable, the validator warns that it "changes state and can be farmed."

MEMORY RECALLS ("the character remembers what you said")

Flags gate content but don't feel remembered. A memory log fixes that: a choice can carry a memoryLog tag, and a dialogue beat can "recall" it — so an NPC echoes this specific earlier choice instead of a generic flag-gated line. A beat (or a gate) can ALSO be keyed on a FLAG — "she remembers what you SAW", not just what you said. See FLAG-KEYED RECALLS below.

HOW TO AUTHOR IT

1. On a choice, set a MEMORY TAG (memoryLog), e.g. "opening_sultry". When the player picks it, the engine stores the player's exact line under that tag (in the game's memories map).
2. On a dialogue beat, add a MEMORY RECALL: a pair of { key → line }, where the key is a memoryLog tag, a flag name, OR an item-present key `item:<id>` (the picker offers all three). The line replaces the beat's normal text when the memory was made / the flag is set / the item was presented. Use the {said} placeholder: for a memory it echoes the player's exact words; for a flag it echoes the flag's NOTE from the Flags tab (or the flag's readable name when no note is written); for an item it echoes the item's NAME ("The Photograph"). When several keys match, the first in insertion order wins.
3. Optionally, gate the beat with a "Remembers" rule (requiresMemories) so it only appears at all once the memory was made or the flag is set — the strongest form of "you remember that I did X."

FLAG-KEYED RECALLS

The recall key and the "Remembers" gate both accept flag names:

- BEAT: recalls: { saw_the_body: "You saw it. {said}" } — plays when the flag is set; {said} becomes the note ("You saw the body in the study."). Write the note like a sentence fragment the line can swallow.
- GATE: requiresMemories: ["saw_the_body"] — passes on the memory OR the flag, so one rule covers both "said it" and "saw it".

In the Test tab these are separate rows: FLAGS toggles real flags (which is what flag-keyed recalls and gates respond to) and Memories toggles memoryLog tags. The validator warns (never errors) when a recall or gate references a key that is neither a memoryLog tag nor any flag in the story.

WHERE IT SHOWS UP

- The recalled line is used for the typewriter text AND for the Web Speech / voice narration (so the echo is spoken too).
- recalled lines are resolved at render time; the beat's own text remains the fallback when no matching memory exists.
- Memory is part of the saved game, so an NPC keeps remembering across loads.

STUDIO

- Choices tab: "Memory tag (memoryLog)" input.

- Dialogue tab: "Memories recalled" key/value editor (tag → line), with a
  datalist of known tags; use {said} inside the line.
- Gating editor: "Remembers" rule (requiresMemories) with a tag picker.
- Test tab: a "Memories" chip row so you can simulate which memories are set.

VALIDATOR

The validator warns (never errors) when a recall or a requiresMemories gate
references a tag that no choice sets — catches typos and dead memories.

MEMORY RECALLS ("the character remembers what you said") — see also Section 10.

"X WILL REMEMBER THAT" CARD (impact card)

A Choice can carry an optional rememberLine. After picking such a choice the
game shows a full-screen impact card that fades in, holds, and fades out
(e.g. "Lydia will remember that."). It is immediate, explicit feedback that
the choice mattered — the classic impact-card beat.
- Set it in Studio (Choices tab → "\"X will remember that\" card").
- Works on both navigating choices and conversational replies.

LETTERBOX & ENTRY TRANSITION (cinematic polish)

A Scene can opt into cinematic presentation:
- LETTERBOX (letterbox): black bars at the top and bottom of the frame.
- TRANSITION (transition): how the scene enters over the previous one —
  "fade" (dissolve), "fade-black" (dip to black then back), or "cut" (instant).
- TRANSITION SPEED (transitionSpeed): seconds for the entry transition.
Set these in Studio (Scene tab → "Letterbox & transition").

QUICK-TIME PROMPTS (action beats with a hard deadline)

A quick-time prompt is a choice with a hard deadline (timeLimitSeconds) that
turns into a tense action beat: a big on-screen prompt, a countdown ring
around the choice buttons, and — as the clock runs low — an optional heartbeat
sting and a swap to SHAKY CAM for the duration of the choice option.

AUTHORING A QUICK-TIME PROMPT (Choices tab → "Quick-time prompt" toggle)

- Set a TIMER (timeLimitSeconds) — the hard deadline.
- BIG PROMPT (qtePrompt) — the on-screen text (defaults to the choice text).
- HEARTBEAT (qteHeartbeat) — plays a low two-beat pulse once the timer
  crosses ~30% remaining.
- SHAKY CAM (qteShakyCam) — the scene's cinematic is swapped to SHAKY CAM for
  as long as a QTE choice is below ~30% remaining.
- TIMED-OUT TARGET (timedOutTargetSceneId) — where to route when the window
  expires with nothing picked. This is the "you hesitated" consequence path;
  it is usually a consequence scene, NOT necessarily a game-over. Defaults to
  the choice's own target.
- TIMED-OUT CONSEQUENCES (timedOutConsequences) — effects applied on expiry
  (e.g. set a "hesitated" flag, add Heat). This is what lets downstream
  beats/choices react to the silence *specifically*, not just to the absence
  of a pick.

HOW IT PLAYS

- The chosen quick-time option navigates to its normal target (or answers in
  place if it's a reply) and applies its consequences, exactly like a normal
  choice — if you beat the clock.
- Below ~30% remaining the optional heartbeat sting sounds (once per window,
  on the tick that first crosses the threshold) and, if enabled, the camera
  swaps to SHAKY CAM for the rest of the window — the swap works whether or
  not the scene already has its own cinematic configured.

- If the timer expires with nothing picked, the choice is NOT silently removed: it applies timedOutConsequences and routes to timedOutTargetSceneId (or the choice's own target). The miss path is a distinct, authorable consequence — the "miss but survive" pattern.
- The clock PAUSES while the ☰ Menu is open and resumes from the paused value — a player weighing a save or a quit can't be routed down the "you hesitated" path behind the menu.

STUDIO

- Choices tab: the "Quick-time prompt" toggle reveals a Timer (seconds) field right in the QTE panel (so the deadline is impossible to miss), plus prompt text, heartbeat, shaky-cam, timed-out target dropdown, and timed-out flag + Heat fields.
- Turning on "Quick-time prompt" for a choice with no timer auto-assigns a 5-second default, so a QTE always has a live countdown ring + banner.
- The Studio canvas draws the timed-out target as a separate "(timed out)" edge, so the consequence branch is visible.
- In-game the countdown ring depletes around the choice button and keeps the choice number visible in its centre; the "□ Ns" readout sits to the right.
- Validator warns if a quick-time choice has no timer, and errors if its timed-out target doesn't exist.

QUICK-TIME DEMO CASE

The included "Quick-Time Demo" case (id: qte-demo) is a tiny smoke-test for this feature. Open Studio, pick it, and hit "Play draft": you'll see the big "STOP HIM!" banner, the numbered countdown ring (number no longer masked out), the □ read-out, the heartbeat + shaking-cam as the clock runs low, and — if you let the clock expire — the "you hesitated" consequence scene with +2 Heat instead of a silent loss.

TYPE FIELDS
---
quickTime? · qtePrompt? · qteHeartbeat? · qteShakyCam? · timedOutTargetSceneId? · timedOutConsequences? (all on a Choice)

SECTION 8 — METERS, TRUST & HEAT

FOUR METERS (0..10)

CHARM	smooth-talking / beguiling (internal key: "Sultry").
INTIMIDATE	force / pressure (internal key: "Aggressive").
DEDUCE	deduction / insight (internal key: "Clever").
HEAT	suspicion — being too pushy, wrong accusations, failed puzzles (internal key: "Suspicion").

The player-facing labels are Charm / Intimidate / Deduce / Heat. The internal keys (used by story data, gating and saved games) remain "Sultry" / "Aggressive" / "Clever" / "Suspicion" — so existing cases and saves keep working; only the labels authors and players see have changed.
Every meter is clamped to 0..10. Choices tagged with an attitude nudge the matching meter via their consequences. Meters drive unlockable perks and the heat trapdoor.

TRUST (per character, -5..+5)

Starts from the character's trustMeter (core usually 0). Raised by Charm/Deduce replies, lowered by Intimidate ones. Some dialogue beats and choices only appear when a character trusts you enough (minTrust).

```
RELATIONSHIP LEDGER (the "Who trusts / fears you" journal)
----------------------------------------------------------
A per-character journal. Tap the "□ Relationships" button
(top-right HUD, next to Case File / Corkboard) to open it. It reads the
player's CURRENT trust in each met character plus the key flags that mention
them, and renders a one-line opinion:
    "Vera trusts you."      trust +2.. | "Vera trusts you completely."  +4+
    "Vera is warming to you."   +1   | "Vera is wary of you."          -1
    "Vera resents you."      -2..   | "Vera fears you."               -4..
    "Vera is still reading you."   neutral
- The opinion is DERIVED from state (characterTrust + activeFlags), so it
  persists across saves and reloads with no extra data to store.
- Flags that reference a character (by id or first-name slug, e.g.
  "contract_caleb" → Caleb) are shown as chips on that character's card, so
  authors can see at a glance which relationship beats have fired.
- Met characters appear in full; unmet cast show as "Not yet met".
- The same opinions also appear as a compact "Relationships" row in the Case
  Progress panel, and live in the Studio Test/Playtester tab (as you toggle
  trust and flags).

PERKS (auto-unlocked by meter thresholds)
-----------------------------------------
VELVET TONGUE   Charm 4+    positive trust gains get +1 (WIRED: applied when
                             a choice grants trust)
IRON NERVE      Intimidate 4+  hard-nosed (Intimidate-tagged) choices stop
                             costing Heat — while active, the Heat delta of an
                             Intimidate choice is dropped to 0 (WIRED)
SHARP EYE       Deduce 4+   hotspots get a shimmer ring; wrong corkboard
                             guesses draw no Heat (WIRED: suppresses the
                             bad-link cost)
HEAT            Heat 6+     positive trust gains are halved; wrong corkboard
                             links cost 2 instead of 1 (WIRED; also shows a
                             pulsing "HEAT" label in Case Progress)
All four perks are surfaced as toasts when they unlock and are tracked in the
Case Progress panel. Velvet Tongue, Iron Nerve, Sharp Eye and HEAT all have
real, wired effects.

TESTING THE PERKS
-----------------
The included "The Perk Range" case is a four-station proving ground built to
confirm each perk in isolation: a bar that pumps each meter, a double-check
station for Velvet Tongue (trust +1 -> +2) and Iron Nerve (Intimidate choice
stops costing Heat), a corkboard station for Sharp Eye (bad guess costs
nothing), and a crowd station that drives Heat to 6 for HEAT. Use it to
verify a perk's behaviour without hunting for a scenario in a real case.

HEAT TRAPDOOR
-------------
If the Case File sets a heatLimit and a heatFailSceneId, then the MOMENT
Heat reaches heatLimit the engine sounds a sting, shows "Too much heat.
The city closes in." and sends the player to the bust scene — including
from meter gains that don't change scene (reply choices, pattern-puzzle
outcomes, on-enter effects), which previously waited for the next scene
change. The bust never fires while the player is already standing on an
ending (so it can't race a simultaneous victory navigation) and it fires
once per run. The Case File panel shows a live "Heat before bust" readout.
In a season the bust scene must sit inside the CURRENT episode.

UNLOCKING "CASE CRACKED"
------------------------
When the player has set every deduction connection on the corkboard, the
engine auto-fires a "CASE CRACKED" toast, grants +1 Deduce and sets the
implicit "case_cracked" flag — a handy gate for the final accusation.
```

```
================================================================================
   SECTION 9  —  HOTSPOT ACTIONS (WHAT A HOTSPOT CAN ACTUALLY DO)
================================================================================

   A hotspot is a clickable region on a scene's background. Its action is a
   single effect, chosen from:

   goToScene     navigate to targetSceneId. (Needs a target.)
   giveItem      grant itemGranted to the satchel. (Triggers a pickup sound and
                 a "Picked up" toast; if the item is a clue and the case has a
                 corkboard, it nudges the player to open it.)
   setFlag       activate flagSet.
   showText      display action.text — as an EXAMINATION CARD: white text on
                 a near-black card above the dialogue panel, titled with the
                 zone's label, holding until clicked (or ~7s). Locked zones
                 get the same card in red for their fail message. Note:
                 showText can also carry a flagSet/meterChange, in which case
                 those apply too (via the action runner).
   meterChange   apply action.meterChange (one meter, one delta).
   openBoard     open the deduction corkboard modal.
   useItem       consume itemConsumed and then apply the rest (like a "unlock
                 with key" — you consume the item to get through, or to trigger
                 the action). Needs a target scene.

   CURSOR TYPE  (feedback in-game)
   -------------
   look (magnifier) / take (grab) / talk (speech) / use (gear) / go (arrow).

   REPEATABLE
   ----------
   If true, the hotspot stays clickable every visit. If false, a state-changing
   hotspot (one that grants an item, sets a flag, or moves) is consumed after one
   use and hidden. A plain showText hotspot that carries no flag or meter
   change is never consumed — it stays clickable because it changes nothing.
   Never mark a reward hotspot repeatable — the validator warns about replaying
   rewards/puzzles.

   SHOW ITEM IMAGE
   ---------------
   When a giveItem hotspot has showItemImage on and the item has artwork, the
   granted item is drawn inside the zone so the player literally sees the object
   sitting in the scene before clicking it (a strong usability win).

   CUSTOM FAIL MESSAGE
   -------------------
   Replaces the generic "That's locked up tighter than a nun's diary." Usually
   best paired with a gating rule so the player learns WHY it's locked.

   GATING ON A HOTSPOT
   -------------------
   A hotspot can be gated so it only works once you're holding an item or have a
   flag (e.g. "use the key on the door"). When locked, it shows the hint
   (or your custom fail message) instead.

   EXAMINE (hold to inspect)
   -------------------------
   A giveItem or showText hotspot can be an EXAMINE zone (Hotspots tab →
   "□ EXAMINE — hold to inspect"). Instead of a click, the player HOLDS the
   zone:
     - The scene CAMERA ZOOMS IN on the hotspot bounds (a clamped CSS
       magnification, ~1.6×–3.5×) over ~0.55s.
```

  - An "EXAMINING — <label>" box pops up with a shimmering progress meter that fills over the authored hold (examineSeconds, default 2.5, range 1–6s). Releasing the pointer early ABORTS the examine (no reward, the zone stays).
  - On completion the normal action fires: giveItem = the object is picked up (pickup sound + "Picked up" toast); showText = the description shows as the examination card. The zone is then consumed (hidden).

AUTHORING AN INVESTIGATION SCENE:
  - Use a LARGE, DETAILED background image. The zoom is a CSS magnification — a small or low-res plate blurs when the camera pushes in. A full-bleed, high-res "last order" plate (a counter with a cold coffee, a watch, a note) is the reference: The Neon Diner's "Mabel's Station".
  - Give each examine hotspot a flagSet, and write a narrator BEAT gated on that flag (conditionRule: requiresFlags) to explain the find. The beat plays as the next line the moment the examine completes.
  - Vary the hold (2–3s) so each examination has its own weight; a longer hold for the payoff object, a shorter one for flavor.
  - A locked examine zone behaves like any locked zone (fail card, no zoom).

END-OF-SCENE INVESTIGATION NUDGE

When a scene's dialogue has run out and there are still SEARCHABLE hotspots left in it (anything that isn't a pure exit), the game tells the player — three ways at once:
  1. a toast when the choices appear: "□ Before you move on — N thing(s) to examine in this scene.";
  2. the searchable zones themselves BREATHE GOLD on the scene art (a soft swelling ring) — the picture literally points at the things worth examining;
  3. a pulsing gold banner in the action list with a one-click "□ Examine '…'" button that examines the first unexamined thing.

It clears the moment the player searches anything (even a locked attempt counts as searching), picks a choice, leaves, or dismisses it, and it never shows on endings, credits or cutscene nodes. Pure goToScene exits don't count as searchable and never glow — the nudge is about looking around, not moving on. Suppressed under reduced-motion (the ring holds steady instead of breathing).

IN-PLACE MINI-GAMES

Because hotspots that grant items/flags keep the player put, people author lockpicks, safes and ciphers as in-place hotspots rather than navigation. Only goToScene and useItem strictly require a target scene; a giveItem/setFlag/showText hotspot legitimately stays in place.

PATTERN PUZZLE (action type "patternPuzzle")

A ledger mini-game: the player flags the rows that share a hidden pattern. Add a hotspot of action type "□ Pattern puzzle", then author its config in the Hotspots tab:

PUZZLE TITLE / INSTRUCTION	the header the player sees.
FLAG COUNT	how many rows to flag — it AUTO-SYNCS to the number of rows you tick as "answer", because a correct solve must flag every answer row. (In play the answer count is the source of truth, so a mismatched targetCount can never make the puzzle unsolvable.)
ROWS	one row per ledger line: date / description / amount, plus "Part of the pattern (answer)".
PAYOFF TEXT	shown when cracked.
ON SOLVE	reward + route — grant an item, set flags, a Deduce delta, and optionally GO TO a scene.

ON WRONG	message, consequences + route — a wrong-guess MESSAGE (shown on a retryable miss), a Heat delta, flags, and optionally GO TO a scene.
ALLOW RETRY	on (default): a wrong guess flashes the message and the player tries again. OFF: a wrong guess is narratively final — it applies the wrong consequences and routes to the wrong scene.

HOW THE PUZZLE BRANCHES THE GRAPH

A pattern puzzle is a real branch, not a dead-end: its solve route (targetSceneId) and fail route (wrongTargetSceneId) are treated as scene edges, so the Studio canvas draws them and the validator counts them as exits (a puzzle-only scene is NOT flagged as having "no exits"). The Solve and Fail consequences (rewardConsequences / wrongSubmissionConsequences) are full Consequences objects — items, flags, meters, trust — applied at the moment of the solve or the final wrong answer.

PATTERN PUZZLE DEMO CASE

The included "Pattern Puzzle Demo" case (id: pattern-puzzle-demo) is a tiny smoke-test. Open Studio, pick it, hit "Play draft", and from one screen you can exercise every path:

- "Open the bank ledger" — a RETRY puzzle: flag the three ~$10k withdrawals to the same bank to crack it (wrong guesses flash a hint, then retry); solving grants a clue +1 Deduce and routes to the gold ending.
- "Decode the cipher note" — a ONE-SHOT puzzle (retry OFF): solve it to route to the gold ending, or get it wrong and there's no second chance — +2 Heat and a route straight to the game-over ending.

STAGE 2D — A WALKABLE SCENE (controllable character)

A scene can be a STAGE the player walks around instead of a static plate they click. Turn it on in the Scene tab under "□ Stage 2D — walkable character". Fields (scene.stage2d):

· playerImage	the character sprite: an image url / asset: reference, or an emoji glyph written "emoji:□" (default).
· spawn	{ x, y } spawn point in scene percentages (default bottom-centre, x:50 y:86).
· speed	walk speed as a % of scene width per second (default 18).
· reach	how close (scene-% of width) the character must get to a hotspot centre to fire it (default 3.5).
· keyboard	allow arrow / WASD movement as well as click-to-walk (default true).

In play the player clicks the ground (a target ripple shows) or steers with the keys. Legacy imports become Stage nodes: automatic walk-over behaviour depends on hotspotTrigger (Section 19), not just proximity. Explicit clicks use the SAME actions and gates — give-item, showText, openBoard and goToScene. Locked markers answer with their fail message when explicitly selected. It is purely a movement / presentation layer: the state model, choices, beats and consequences are unchanged. Coordinates are scene percentages (x = % of width, y = % of height); the vertical step is aspect-corrected so stride speed is uniform across a wide plate. Author your hotspots where the player should walk to something. Try the office in Demo Lab: Press the Lie.

SECTION 10 — CHOICES (NAVIGATION, REPLIES, HIDDEN, GROUPS, TIMERS)

A choice is the primary navigation + state-change unit. Every choice has a target scene OR is a reply.

PLAIN NAVIGATION

A normal choice navigates to targetSceneId. Good for "Go to the office", "Head to the club", "Accuse ...".

ATTITUDE TAG

Neutral / Charm / Intimidate / Deduce (the internal tag values remain "Sultry"/"Aggressive"/"Clever"; they display as Charm/Intimidate/Deduce). Tags colour the button and are the conventional way to pump the matching meter (put the corresponding meter delta in the choice's consequences).

CONVERSATIONAL REPLY (reply = true)

A reply does NOT navigate. It stays in the scene, applies its consequences (flags, meters, trust, items) immediately, and then the scene re-evaluates its dialogue beats. The intended pattern:
    1. Write a reply choice that sets a flag (e.g. "gentle_start").
    2. Write a dialogue beat gated on that flag (conditionRule
       { requiresFlags: ["gentle_start"] }).
    3. When the player picks the reply, the beat becomes the response.
Replies are one-shot by default and can share an exclusive group so only one opening answer survives.

HIDDEN (hidden = true)

A hidden choice is not rendered at all until its gate passes. Use it for sequential dialogue where you don't want earlier questions to spoil later ones. (The alternative — leaving it visible but disabled — shows a hint.) A hidden choice with a gate naturally "reveals" itself when the player has done the prerequisite.

TIMER (timeLimitSeconds)

Gives the choice a countdown; when it hits 0 the choice is removed and can no longer be picked (it's recorded as expired). Useful for pressure beats.

EXCLUSIVE GROUP

Choices sharing the same exclusiveGroup are mutually exclusive: picking one marks ALL of them as used, so the alternatives vanish. This is the standard way to author "choose your opening approach" without dead choices lingering.

REPEATABLE (default is derived)

If you don't set it:
  - replies are one-shot (they advance dialogue; allowing re-fire would loop),
  - a choice with consequences (state change) is one-shot,
  - a pure navigation choice is repeatable (safe to reuse for backtracking).
Set isRepeatable explicitly only for intentional loops (e.g. a menu hub), and avoid pairing it with rewards/meters/trust — the validator warns it can be farmed.

PRESENT ITEM ("show the photograph")

A choice can PRESENT an item to the character in the scene — the noir verb. On a choice:
  - PRESENT ITEM: pick the item the choice shows. The choice becomes
    un-pickable (and, if hidden, stays hidden) until the player HOLDS the
    item — the engine adds it to the effective gate, and the locked hint

    reads "needs The Photograph". No authored requiresItems needed.
  - HAND IT OVER (consumePresentedItem): when on, presenting REMOVES the
    item from the satchel (they keep it — "hand over the ledger"). Off
    (default) = you flash it and keep it. The choice being one-shot already
    prevents showing the same item twice for two reactions.
Presenting records an `item:<id>` memory holding the item's NAME, so:
  - a later beat can RECALL it: recalls: { "item:photo": "You flashed
    {said}." } where {said} = "The Photograph";
  - a gate can REMEMBER it: requiresMemories: ["item:photo"].
The item: keys appear in the Studio pickers next to memoryLog tags and
flags. In The Neon Diner, "Show him the photograph" presents the photo,
Ruiz's Act-Two beat is keyed on that presentation, and both evidence
endings HAND OVER their item (consumePresentedItem), so each ending locks
itself if you never built the item it needs.

INTERROGATION (composure pressure + urgency)

Talking is the core loop — give it stakes AND a clock. A scene can run an
INTERROGATION on one of its characters (Scene tab → ▯ Interrogation):
  - COMPOSURE (default 3): the suspect's guard. Every PRESSING reply —
    any reply choice tagged Sultry / Aggressive / Clever — drops it by
    one (a press thunk + gauge jolt + a short reaction line). Neutral
    replies (plain questions, silence, small talk) are free.
  - AT ZERO, TRUST DECIDES (this is the risk): if the suspect's trust is
    at/above the CRACK THRESHOLD (default 0) they CRACK — the ★ BROKE
    stamp slams in, the broke beat plays as the next line, the broke flag
    is set, and the broke swing (trust/heat) lands. Below the threshold
    they CLAMP — the ✖ CLAMPED stamp, the clamped beat, the clamp flag,
    and the backfire swing (usually heat+, trust−).
  - HESITATION HAS A COST (the urgency): after each press, a "steadying"
    window runs for STEADY SECONDS (default 7, range 3–12). A faint line
    creeps across the gauge the whole time — the moment is passing. If
    it completes before your next press, the suspect STEADIES: composure
    +1 (once per scene), a soft exhale, "They steady themselves." Keep
    the pressure on and the window never completes.
  - LIVE STAKES: the gauge is a pressure bar that shifts colour
    (calm → strained → on the edge) with a stage label, and a live
    prediction under it: "◈ break them now — they'll talk" (green) or
    "…they'll shut down" (red), which flips as you build or burn trust.
    At one composure (critical) the room closes in with a pulsing red
    vignette and the suspect's heartbeat is audible.
Composure persists to the save, so a mid-interrogation save keeps its
state; presses are one-shot so you can't re-make a line to grind.
AUTHORING:
  - Give the scene at least composure-many pressing replies, or the
    validator warns composure can never break.
  - Design the WARM PATH (trust-positive lines first, then press to zero
    → crack) and the COLD PATH (press cold → clamp). The Neon Diner's
    booth is the reference: the neutral ask is +1 trust and free, the
    Sultry photo is a press that also builds trust, the hard Aggressive
    presses cost trust, and a 6s steady window with Vince-specific
    reactions — so who you press, in what order, and how fast you press
    all decide the outcome.
  - Set STEADY SECONDS shorter for a twitchy, urgent suspect; longer for
    a stoic one. Add pressReactions[] for the suspect's own escalating
    lines as they're wound (one per level, from first press to the edge).
  - Gate the PAYOFF on the broke flag, and the FALLOUT on the clamp flag.
    Both flags are listed in the ▯ Flags tab (note them there) and are
    treated as settable by the validator.

AVAILABILITY & GATING HINT

A choice shows "locked" and a hint (e.g. "needs The Dead Man's Note & ...")

when its gate isn't met. Hidden choices bypass this and simply don't render. If a gated choice's requirement is a corkboard deduction flag, the game marks it "[board unlock]".

SECTION 11 — DIALOGUE BEATS (SPEAKERS, BACKGROUNDS, VOICE, CONDITION RULES)

A dialogue beat is a single line shown after the scene prose (or interleaved with it). It is the written script of a scene.

SPEAKER

A character, or the reserved "narrator" (V.O.) voice. Speaker is what drives the portrait shown (if the speaker is a character) and whose voice is used for text-to-speech. A beat whose speaker is a character the player hasn't met yet still counts as "meeting" them (which makes them appear on the corkboard).

EXPRESSION

A free-text emotional tag, shown as a small "(expression)" label under the portrait. It's also a cheap way to mark voice-over lines: an expression that starts with "V.O." is rendered as first-person narration.

PER-BEAT REPLACEMENT BACKGROUND

Gives this one beat its own image (different from the scene background). It transitions in and out when the beat starts/ends. Transitions available:
  cut / fade / crossfade / fade-black / slide-left / slide-right / iris-in.
You control the transition speed too. A beat's replacement image has its own cinematic settings (Section 13). Author "insert shots" this way.

VOICE MP3

A per-line recorded audio that replaces TTS for that beat. Useful when you have a real voice take; leave blank to use Web Speech.

SILENT

Text only, no narration at all. Use for busy screens or for lines you want the player to read without audio.

CONDITION RULE (Gating)

A gate that decides whether the beat renders at all. The classic use is to pair it with reply choices (a reposed beat gated on the flag a reply set), but you can also gate on items, meters or trust. Because beats re-evaluate whenever world state changes, a beat that unlocks mid-scene appears as soon as its precondition is satisfied — no reload needed.

PROSE & NARRATION

The scene's prose is the "opening paragraph" and is auto-narrated on first entry (unless autoNarrate is off). If you set noTypewriter, the prose is shown instantly rather than type-written. narrationAudioUrl replaces Web Speech for the prose with a recorded MP3.

PRESS THE LIE (the contradiction verb)

A beat can carry a LIE — a false statement the player must disprove with evidence. Author it on the beat in the Dialogue tab via the
"□ Press the Lie (contradiction)" panel.

```
   · claim                 what the player challenges (defaults to the beat's
                           own line). Shown above the evidence bag.
   · evidenceItemIds       the item id(s) that PROVE the claim false. Usually
                           exactly ONE clue. Any other held item is a bluff.
   · caughtBeat            the reaction that plays when the right evidence is
                           presented. It is spliced in as the next line and is
                           gated on contradictedFlag, so it only fires on a win.
   · contradictedFlag      set when the lie breaks; gate the aftermath
                           (choices/beats) on it.
   · contradictedConsequences  the payoff swing (trust up, heat down, a flag).
   · wrongBeat / wrongConsequences  what the speaker says and what it costs
                           when the wrong evidence is shown (usually heat up /
                           trust down).
   · wrongEvidenceLimit    misses before the claim shuts for the scene
                           (default 2).
   · composureDamage       in an INTERROGATION scene, a proven lie also does
                           this much composure damage (2+ cracks the suspect,
                           running the normal crack/clamp-by-trust resolution).
   · rememberLine          optional impact card on a successful catch.

   Play: while the line is on screen a pulsing "□ Press them on that" button
   opens the satchel; the player presents evidence. Right clue → caughtBeat +
   flag + swing. Wrong clue → wrongBeat + cost, and after the limit the button
   goes dead ("they shut you down"). ALWAYS give a failed press an ungated exit
   choice so a shut-down claim can't soft-lock. The validator checks that the
   evidence exists and is obtainable, that the caught/wrong beats have speakers,
   and that the caught beat is gated on the caught flag. Worked example:
   Demo Lab: Press the Lie (the pawn ticket breaks the alibi; the charred
   matchbook is a red herring; the bar tab cracks the interrogation).


================================================================================
   SECTION 12  —  THE DEDUCTION CORKBOARD
================================================================================

   The corkboard is the "solve the case" visualization. You only need to build it
   if you want the player to literally connect clues to suspects.

   SETUP
   -----
   - A case only has a corkboard once you define a deduction object (section
     5, DEDUCTIONBOARD).
   - SUSPECTS: mark characters with "Show this met character on corkboard". They
     become suspect nodes once the player has met them.
   - CONNECTIONS: a list of {a, b, flagSet, revelation}. Each endpoint is either
     a clue item id or a character id; one should be a clue and one a suspect
     (the engine only accepts clue-to-suspect strings). When the correct link is
     drawn it sets flagSet and shows the revelation.

   HOW PLAYERS SOLVE IT
   --------------------
   The board shows clue cards (from the satchel's isClue items) and met suspects.
   The player drags from a card's red pin onto another card. If the pairing is one
   of your connections that hasn't been solved, it's marked solved (sets flagSet,
   shows the revelation, +1 Deduce, sound sting). Wrong guesses add Heat (or
   are silently ignored if the player has the Sharp Eye perk; cost x2 if they have
   the HEAT perk).

   WHEN EVERYTHING IS SOLVED
   -------------------------
   "CASE CRACKED" fires: +1 Deduce, sets the implicit case_cracked flag. This is
   the natural gate for the gold-ending accusation. Gate the final choices on the
```

individual connection flags (e.g. requiresFlags ["note_malveaux","ledger_malveaux", "photo_alibi"]).

AUTOMATED REDUNDANCY CHECKS

The validator ensures each connection's endpoints exist and that its flagSet is actually set by a connection (otherwise the gate can never be satisfied).

SECTION 13 — CINEMATIC CONTROLS (CAMERAS, FOREGROUND, LIGHTING, PER-BEAT)

Cinematic settings "grade" a single image: either the scene background or a dialogue beat's replacement background. Pick WHICH image to grade with the "Apply cinematic to" dropdown, then configure camera / foreground / lighting. The inspector shows a live animated preview so you can see the motion. Every image preview here is boxed to a 16:9 aspect so it matches the game viewport, and the camera/lighting/foreground layers all render at that proportion (instead of a short fixed-height strip that cropped or distorted the art).

CAMERA MODES

SLOW PAN	a gentle drift. Pan direction (left/right) and drift distance (% of width). 0% drift keeps the soft push-in only.
ZOOM TO SUBJECT	zooms in on the focal point then holds. Set the zoom duration.
SHAKY CAM	handheld jitter: shake amount (intensity → jitter px), roll (camera twist, degrees), jitter speed (lower = faster, more panicked).
RAIL / DOLLY	slides laterally across a WIDE image: direction, travel distance (%), vertical drift (turns it into a diagonal/crane pan), and duration per pass.
DOLLY ZOOM	the Hitchcock vertigo push-pull: background zoom amount, cycle time, direction (push/pull/both), and background blur (px). Pair with a foreground subject so the subject stays sharp (that's what sells the vertigo). THE VERTIGO RECIPE: set direction "pull" (the background eases from the zoom amount down to 1× — the room falls away) while the FOREGROUND runs the opposite move (foregroundZoom > 1 pushes in on the anchor). The two opposite moves stretch the space between the layers. Add keyframes (the keyframe editor) and the subject plays its poses once, then holds the last frame while the vertigo keeps running.
STATIC	locked frame.

FOCAL POINT

A CSS background-position string like "50% 50%" (centre) or "60% 40%" (upper-right). Set it by clicking a point on the live preview, or type the percentage. For pan/zoom-hold, the focal point drives the zoom target.

FOREGROUND OVERLAY

NOTE: this is the PER-SHOT subject layer for standard scenes and dialogue beats (keyframes, zoom). WALKABLE STAGE rooms use a DIFFERENT feature — scene.cinematic.foregroundExtras with occluders, parallax, scrolling and alpha masks — documented in SECTION 20. Same word, different feature.
Layers a fixed image (character, prop or frame) over the scene in front of any camera motion, so the subject stays put while the world moves.
- image (or pick an uploaded one; transparent PNG works best)
- scale and fit: contain / cover / fill / native

- POSITION: drag the subject around, or set its offset from centre in pixels
- FOCUS POINT: click the spot ON THE SUBJECT to define the zoom focus (anchor). It stays pinned while the foreground zoom plays; dragging moves the subject.
- FOREGROUND ZOOM: a slow push-in or pull-out focused on the anchor. Use the quick-set buttons — "Pull out" (0.80x, zoom away from the anchor), "Push in" (1.25x, zoom toward it), or "Off" (1x, base scale) — or drag the slider (0.5x–3x; below 1x pulls out, above 1x pushes in). The zoom animates from the base scale to base × multiplier and holds, and replays whenever the value or duration changes.
- The foreground previews and the cinematic preview render at a 16:9 aspect so the background and subject display at the same proportion they'll have in-game (no squashed/cropped strip).
- KEYFRAME ANIMATION: a sequence of images played in order.
    dissolve = cross-fade between poses (smooth, dreamlike),
    frames   = hard sprite swap (the Ace Attorney / Hotel Dusk look).
  Set frame time, fade time (dissolve only), and loops (loop forever, or play N times and hold the last frame). Use the same base canvas so poses don't drift.

LIGHTING

- Dim amount: overall image brightness 0..0.9.
- Light glows: add point lights (a lamp, window, neon sign). Each light has a colour, size (%), brightness, position (%), and a pulse speed/depth. They glow additively (screen blend) and breathe on a sine wave, and they stay bright even when the image is dimmed (the dim layer sits below the glows).
- Legacy single-light fields (lightSource/lightColor/lightRadius) are merged into the lights array on next edit, so old drafts keep their glow.
- Quick "Cast lens flare from this light" button links a dynamic lens flare directly to any point light source.

DYNAMIC LENS FLARE

Simulates real optical camera artifacts when bright light enters the lens system: anamorphic horizontal laser beams, starburst diffraction spikes, and a multi-spectral chain of chromatic ghost rings along the optical axis.
- OPTICAL PRESETS:
    Anamorphic Blue   CinemaScope / Sci-Fi widescreen blue laser streak with electric cyan reflections
    Golden Hour / Amber Warm sunset & vintage tungsten streetlamp glow with golden streak & amber rings
    Sci-Fi Cyan   Cybernetic cobalt & electric cyan beam with sharp aperture reflections
    Neon Noir   Cyberpunk magenta streak with purple & cyan chromatic reflections
    Vintage 35mm   Spherical lens flare with chromatic rainbow halo, iris rings & starburst rays
    Natural Sun   Soft daylight solar glare with airy pastel rings & delicate diffraction rays
    Interrogation   Blinding high-contrast stark white beam with crisp sharp diffraction starburst
    Emerald Night   Moody noir green / eerie night-vision flare with teal secondary reflections
    Custom   Fully custom-authored optical profile
- POSITION: Click anywhere on the interactive preview to place the flare light source. The optical axis automatically computes and projects across the centre (50%, 50%), orienting every internal ghost reflection and aperture diaphragm disc. Quick buttons allow snapping to Center, Top-Left, Top-Right, or directly linking to any Point Light.
- ANAMORPHIC STREAK: Author the horizontal laser beam intensity, vertical thickness / spread, and angle (-45° to +45°).
- OPTICAL GHOST REFLECTIONS: Configurable ghost intensity, axis spread multiplier (0.5×–2.0×), and detail tier (Subtle 4-element, Standard 8-element with rainbow chromatic halo, or Rich 12-element multi-spectral chain).
- STARBURST DIFFRACTION: Diffraction ray intensity and spike count (4, 6, 8, or 12 rays) with continuous glint/shimmer.
- CORE & CORONA GLOW: Sizing for the brilliant hot white core and radial corona.
- DYNAMIC SHIMMER & CAMERA MOTION REACTIVITY: Shimmer pulse speed, breathing depth, and camera parallax reactivity multiplier (0×–2×). The lens flare dynamically reacts to camera movement, panning, dolly tracking, and handheld camera jitter with physical optical inertia.

PER-BEAT CINEMATIC

If a dialogue beat has its own replacement background, that beat gets its own cinematic settings too (select it in the "Apply cinematic to" dropdown). The beat's camera/lighting/lens flare replace the scene's while the beat plays, then revert. This is how you do a slow push-in on a character's reaction shot.

REDUCED MOTION

If the user prefers reduced motion, all animations collapse to near-zero duration via CSS media query — worth remembering for pacing-sensitive scenes.

SECTION 14 — VALIDATOR, REPAIR, PLAY & PUBLISH

VALIDATOR (Continuity Department)

The validator presents issues in two severities:
- ERROR (red) blocks Publish and Play draft.
- WARNING advisory; does not block, but almost always worth fixing.

Click an error to jump to its scene and centre the camera.

WHAT IT CHECKS

- duplicate scene / item / character ids
- a start scene id that doesn't exist
- dangling links to missing scenes
- unreachable scenes (from the start)
- a non-ending scene with no exits (dead end)
- a non-ending scene whose every exit is gated (softlock risk)
- an ending missing endingData
- gating that requires a non-existent item
- a choice with no target that isn't a reply
- a reply choice that ALSO has a target (it will navigate instead)
- a repeatable choice that changes state (farmable)
- a choice granting a non-existent item
- a hotspot with invalid/inverted percentage bounds
- a goToScene/useItem hotspot without a target scene
- a repeatable hotspot that changes state or moves scenes
- a dialogue beat whose speaker isn't a real character (reserved "narrator" ok)
- a music-mode scene with no track selected
- no victory ending (ERROR — player can't win)
- no game-over ending (WARNING — no real risk of losing)
- no Case File objective (WARNING)
- a heat-bust scene that doesn't exist, or isn't an ending
- corkboard suspects that aren't characters; connections referencing missing nodes; a connection with no clue endpoint
- recipes referencing missing ingredients/results
- items required by a gate but never granted anywhere (ERROR)
- flags required by a gate but never set by any choice/hotspot/connection (ERROR)

CRAFT CHECKS (advisory warnings — the story can be walked but never LANDS)
- a flag set but never read by a gate, beat recall, recap or flag note — the player's action changed nothing (the automated backward-design ledger)
- a flag cleared but never set
- a memoryLog tag nobody recalls or gates on ("the character remembers" — then never does). Silent in episodic seasons (the end card lists them)
- a "will remember that" card with no memory tag and no state behind it
- a press-the-lie beat with no/unknown evidence, no caught flag, a missing speaker on its caught/wrong beat, or a caught beat not gated on its flag

- a quick-time choice whose timeout lands on the same scene/reply with no timeout consequences (a fake deadline)
- two visible choices with the same target, attitude, effects and no memory (the same decision twice)
- a run of 4+ reachable scenes with no choice, hotspot or interrogation (a passive corridor) — collapse it or ask something mid-run

REPAIR (auto-fix)

One click fixes the common strands:
- resets a missing start scene to the first scene
- strips gate requirements that reference deleted items
- re-points dangling choices/hotspots to the start scene
- removes reward items that don't exist
- makes reward hotspots one-shot
- clears granted/consumed items that don't exist
- adds endingData to flagged endings
- adds a guaranteed open exit to any dead-end scene
- clears broken recipes

It reports how many fixes it made.

PLAY DRAFT

Runs the draft in the game loop. Blocked while edits have hard errors. Use the Test/Playtester tab to verify gates quickly first, then Play draft for the real end-to-end run.

PUBLISH

Compresses the story (storing compact asset references) and pushes it to the newsstand. Blocked for blank cases and for hard errors. If storage is full it reclaims stale backups and retries once. After publishing, the newsstand shows exactly this build.

SECTION 15 — EXPORT, IMPORT & SHARING

EXPORT

Bundles the story plus every embedded media asset into one .reel.json. Download it or copy the JSON. Because media is included, the file travels complete — it is the way to back up a case or hand it to another author. The media pipeline covers every asset-bearing field: scene backgrounds, music, narration, voice lines, video cutscenes, foreground overlays, keyframe animations, per-beat backgrounds, character avatars, talking frames AND resting poses, item icons, act title cards, episode title cards / posters / card music / recap stills, the cover image, dialogue-box backdrops, and credits montages.

IMPORT

Pick a .reel.json; Studio restores scenes, choices, characters, items and every embedded asset, then saves it compacted as your current draft. It rejects files without scenes or a start scene.

SHARE

Encodes the whole story into the URL hash (#reel=base64…). Copy the link; anyone who opens it plays your case. Very large cases are too big for a URL — export instead.

TIP

Export before major refactors. If autosave ever fails on quota, export your case, run Storage to reclaim space, then re-import.

SECTION 16 — KEYBOARD SHORTCUTS

STUDIO

Ctrl+Z	Undo
Ctrl+Shift+Z / Ctrl+Y	Redo
Delete / Backspace	Delete the selected node or edge
Esc	Close context menus / dialogs

IN-GAME (player)

1 - 9	Pick a choice by number
Up / Down W / S	Move choice focus
Space / Enter	Continue — a press while text is still typing FINISHES the line first (no swallowed sentences)
Left arrow	Rewind one line
Shift + Left arrow	Restart the scene
B	Open the corkboard
L	Open the dialogue log (everything said this run)
A	Toggle auto-advance (hands-off reading; remembered)
H	Reveal hotspots — shimmer every clickable zone (3.6s)
M	Mute / unmute (remembered between sessions)
.	Skip current text
Esc	Close / quit
Esc / Enter / Space	on an interstitial (part-break chart, act route map, act/episode title card) — dismiss and continue

QUALITY OF LIFE

AUTO CHECKPOINT	an invisible save is written on every scene arrival; the newsstand's "▶ Resume" and Load → "□ Auto checkpoint" both return to it, so a crash or a closed tab never costs more than the current scene. A brief "✓ Checkpoint" pulse confirms the save landed.
AUTO-ADVANCE	toggle in the ☰ Menu or press A — hands-off reading that advances dialogue on a comfortable timer once a line finishes typing. Stops at every choice so the player still decides. Remembered per player.
SKIP TEXT	toggle in the ☰ Menu — fast-forward mode that types instantly and steps with a brief gap. Great for replays. Remembered per player.
REVEAL HOTSPOTS	press H or use the ☰ Menu — shimmers every clickable zone in cyan for ~3.6s so a stuck player can see what's interactable without spoilers. Also reveals the hotspot labels. Resets on scene change.
EXAMINATION CARD	hotspot results (and locked-zone messages) appear as a card above the dialogue panel, titled with the zone's label, holding until clicked (or ~7s). The card text is also narrated via TTS (respecting mute / auto-narrate settings).
DIALOGUE LOG	every spoken line of the run (prose + beats, with speakers) is recorded as it plays — press L or use ☰ Menu → □ Dialogue log. Skipped text still lands in the record; the log auto-scrolls to the newest line.
ART PREFETCH	while the player reads choices, the browser quietly decodes the target scenes' backgrounds (and the next

```
                      beat's insert shot), so scene cuts swap plates
                      instead of flashing black.
   PERSISTENT MUTE    the sound toggle survives reloads.
   END-OF-GAME FLOWCHART  the route flowchart at an ending shows the FULL
                      journey (every decision across the whole case), not
                      just the segment since the last part break.
   END-OF-PART FLOWCHART  mark a scene with an "End-of-part route chart"
                      name (Scene tab) and, when the player leaves it, the
                      decisions since the previous chart animate across the
                      screen. Esc / Enter / Space dismisses it like the
                      other interstitials; global shortcuts (1-9, Space,
                      B, L, …) stand down while it's up, so they can't
                      fire the old scene's choices behind the overlay.
   END-OF-ACT ROUTE MAP  when a part exit crosses an act boundary, the
                      act you're LEAVING gets a full Detroit-style branch
                      map (every scene, your route lit, the rest grey)
                      before the next act's title card. Same keys dismiss
                      it. Act membership is DERIVED from the graph: an act
                      owns every scene reachable by choice edges, timeout
                      routes and door hotspots from its start scene,
                      stopping at the next act's start. Membership is
                      STRUCTURAL — a door behind a key gate still shows on
                      the act's map even in a run where the player never
                      gets the key — and a scene no act's walk reaches
                      (typically the heat-bust trapdoor) belongs to no act;
                      the validator notes it. That's a note, not an error.
   ROUTE NOTES        both charts end with a "Route notes" list: the flags
                      the stretch just played set (from the decisions'
                      consequences, including QTE timeouts), each with its
                      note from the ⚑ Flags tab — a "what has changed in
                      this case" log.
   CASE PROGRESS      the top-left panel lists meters, perks, suspects,
                      clues and corkboard progress, plus "Story so far —
                      flags set": every active flag with its note (or its
                      readable name). It scrolls when the case outgrows the
                      screen.

   GAMEPLAY UI
   -----------
   The top-right HUD controls are grouped behind a single "☰ Menu" button
   (Subtitles, Case File, Corkboard, Relationships, Dialogue log, Save,
   Load, Quit) so the scene stays clean — click it or press Esc to close.
   SUBTITLES (player option): "Cinematic" is the default — big pulp
   lettering centred over the art. "Classic" switches to compact
   reading-size captions with a gold SPEAKER label on every line, for small
   screens or plain-subtitle readers. The choice is remembered per player. The one you pick (keyboard focus)
is
   marked with a neon ring and a "⏎" hint. Each choice shows a number pill (or
   a QTE countdown ring) and an attitude icon, so you can read Charm /
   Intimidate / Deduce at a glance.

   DYNAMIC GAMEPLAY LAYERS
   -----------------------
   - CINEMATIC LOCATION CARD: on every scene arrival a big pulp-lettering
     lower-third title slides in, holds, and fades out (Wolf Among Us style).
   - HEAT / RISK VIGNETTE: when Heat (Suspicion) is within 2 of a bust — or at
     the HEAT perk threshold — a crimson edge glow pulses around the screen,
     the Warning "Heat — lay low" flashes, and the Heat meter bar pulses in
     Case Progress. Danger reads on screen before the trapdoor fires.
   - CHOICE ENTRANCE & HOVER: choices stagger in and lift slightly on hover so
     picking an option feels tactile.

   DIALOGUE AS SUBTITLES (Wolf Among Us style)
```

In a standard scene the prose and dialogue lines render as cinematic subtitles over the art (not a heavy panel): the speaker's name sits as a small gold label, the line is centred lower-third in a clean serif (Newsreader, medium weight) with a drop shadow, and a subtle gradient scrim darkens the bottom so it reads over any background. The character portrait is no longer drawn — the voice carries it. A tiny row below shows back / continue / speak. Audio narration is independent of the text, so you can play it as pure voiced subtitles. Classic subtitles use the same serif at a smaller reading size with a gold SPEAKER label prefix. The old Special Elite typewriter font has been replaced everywhere with Newsreader (a clean literary serif) for readability. Credits and cutscene nodes keep their own full-screen presentation.

CHOICE STYLING

Choice buttons are shown in clean serif. **Conversational replies** (the ones that answer in place, marked □) get a distinct cyan rounded card, while **navigation choices** keep the attitude-coloured left bar — so dialogue answers look clearly different from "go there" options. A focused choice gets a neon ring + a "⏎" hint; timed (QTE) choices show a countdown ring instead of the number pill.

SECTION 17 — THE FIVE-STEP AUTHORING WORKFLOW

The Creator workflow panel highlights the intended order and lights up each step as it's satisfied.

1. CASE FILE
   Set the objective, then the win and lose conditions. Players read these in-game, so a blank Case File leaves them guessing. Add a heat limit and a bust scene if you want real stakes.

2. PEOPLE
   Create the suspects and supporting cast. Mark who should appear on the corkboard and tune their voices.

3. ITEMS
   Add the clues and props. Mark clues and set any combining recipes. Make sure EVERY item required by a gate is actually granted by a hotspot or a choice's reward.

4. SCENES
   Write the prose, add dialogue beats, scatter hotspots, and author choices. Wire exits by dragging handles. Finish with at least one victory ending AND at least one game-over ending.

5. TEST
   Open the Test tab to fake the world state and confirm each gate is reachable, then Play draft. Fix any red errors, then Publish.

SHIP READINESS PANEL

A one-glance health check: Start scene, Ending scene, No hard errors, Case objective — plus a button to open the Validator.

SECTION 18 — TIPS, STORAGE & TROUBLESHOOTING

```
================================================================================

  AUTHORING TIPS
  --------------
  - Always provide BOTH a victory and a game-over ending. The validator treats
    "no victory" as an error and "no game-over" as a warning.
  - Never gate the only exit of a scene behind a requirement you never grant —
    that's a softlock. The validator catches required-but-ungranted items/flags;
    keep them in sync.
  - Keep state-changing choices and hotspots one-shot. Repeatable + rewards =
    farming.
  - Use exclusiveGroup for "choose your approach" beats so alternatives don't
    linger.
  - Use reply choices + gated dialogue beats to write conversational Q&A instead
    of menu spam.
  - Use the corkboard as the finale: wire clue -> suspect strings, then gate the
    accusation on those deduction flags.
  - Set the focal point by clicking the subject in the foreground preview; it
    stays pinned during the zoom.
  - Check the validator before every Publish.

  STORAGE & AUTOSAVE
  ------------------
  - Autosave runs ~0.8s after edits; if it fails on quota you'll see a warning
    and the Storage button to reclaim stale backups.
  - Media is stored compactly. If storage is genuinely full: Export -> Storage ->
    re-import.
  - PLAYTHROUGH SAVE SLOTS CARRY A SCHEMA STAMP. Saves written before
    versioning existed (no stamp) load fine and are treated as v1. A save from
    a NEWER build is refused rather than mis-loaded, so an old app won't
    corrupt progress made in a new one — reload the app, and the slot comes
    back.

  TROUBLESHOOTING CHEAT SHEET
  ---------------------------
  PUBLISH / PLAY BLOCKED
       Open the Validator and fix or Repair the red errors.
  A SCENE HAS NO WAY FORWARD
       It's a dead end: add a choice/hotspot exit, or mark it as an ending.
  A CHOICE IS LOCKED AND YOU NEVER GET THE ITEM
       It's required but never granted — add a hotspot/choice that grants it.
  CHARACTERS DON'T APPEAR ON THE BOARD
       Mark "Show this met character on corkboard" and make sure the player meets
       them.
  THE CASE IS UNWINNABLE
       Check you have a victory ending and that the final gate's flags are all
       settable via the corkboard/choices.
  MUSIC ISN'T LOOPING
       Set the scene's music mode to "Play a track" (not inherit) and pick a file.
  PROSE SHOWS AS RAW TEXT
       Turn off noTypewriter (or set a narration MP3).
  A BEAT NEVER APPEARS
       Its condition rule isn't satisfied — check the gate and the state that
       sets it.
  FOREGROUND NOT VISIBLE
       Confirm the image loads and you have a non-empty foreground (image or
       frames), then check its position/scale/fit.
  THE PREVIEW HOST IS REJECTED BY VITE
       Ensure dev/preview host allow-lists include your preview origin.

  FINAL NOTE
  ----------
  Repair handles common graph/data errors, not missing story logic, art
```

alignment or inaccessible floor geometry. Export first, then Repair if appropriate, re-check the Validator and perform an actual playthrough.

SECTION 19 — PEOPLE ON STAGE: NPCS, PATROLS & SET MOTION

A stage scene's NPCs (stage.npcs[]) are the room's actors: they stand or patrol the floor, play a SPRITE SHEET (or a static cutout), and can OWN a conversation marker so the character IS the hotspot. Ambient NPCs with no link are pure set dressing — the crowd on the street, the cook at the bar — and cost nothing but a sprite and a route.

SAFE INTERACTION & RETURN ROUTES

Stage -> Interaction exposes hotspotTrigger. Use aim for a Westwood-like point-and-click case: moving through a pickup or a door does NOT activate it. Smart allows ordinary object walk-over but keeps scene exits deliberate. Walkover explicitly opts into legacy automatic activation, including exits. A clicked target owns the whole approach even in smart/walkover rooms: an incidental object cannot intercept it. One arrival fires once. A new direction key or an empty-floor click cancels the old aim and NPC pursuit. A click inside another hotspot is a NEW interaction, not an empty-floor command. Deliberately authored walk-off edge exits still work in aim rooms.

Labels are non-interactive tooltips above a compact marker target, not invisible full-width hitboxes. The container inventory's empty transparent area lets clicks through; its item chips and controls remain interactive. Keep real HUD buttons away from essential markers and test small screens.

Test tab -> Walking room navigation audit -> choose a return destination -> Run navigation audit. This checks fresh-state gates and routes from the default spawn to local exits, pickups and conversations. It uses the same wall-slide step as play, not merely a route's requested endpoint. An endpoint inside a counter is NOT proof that a disk can be picked up. Maps do not count as doors. Click a reported room to fix its approach. After editing, rerun. Also play from every entry spawn and earned state: moving actors, scripts, timed choices, HUD overlap, minigames and changing gates are not exhaustively proved by this static/current-state audit.

Always offer an ungated, repeatable return to an investigation hub. Do not put it behind the same clue required to finish the case. In Blade Runner, stage 12's apartment elevator -> LAPD mainframe and the home control back are two ordinary goToScene hotspots. They work without ESPER evidence and remain available after uploads, conversations, and either Zuben outcome.

HIDDEN WALKER & PHOTOGRAPHIC PRESENTATION

Stage -> Player character -> Hide player sprite sets stage.hidePlayer. For cockpits/close-ups also make controls remote; hiding a sprite does not turn every hotspot into a remote control. Disable keyboard input for a seated shot; do not rely on speed zero to make a distant hotspot usable. Characters -> Pulp paper changes story.pulpPaper. Set false for photography; an existing player's saved paper preference still takes precedence.

THE ACTOR

- id, x, y — spawn position, % of the WIDE world (x 0..worldWidth, y 0..100). Actors live in the room's floor band and take the same depth scaling as the player: higher on the plate = farther = smaller.

```
- sheet / image       a SpriteSheet (idle at rest, walk while patrolling)
                      or a static image / `emoji:` fallback.
- scale, facing       flat scale on top of depth scaling; left/right. Player
                      and NPC transforms scale about their bottom-centre:
                      x/y remain the FEET position even at large HD scales.
- gating              Stage -> Ambient NPCs -> Gating rules controls when
                      an actor exists. Requires/excludes flags/items and
                      the normal meter/trust/memory rules are reusable.
                      Give a linked talk marker the same gate and tick
                      Hotspots -> Hide marker when its gate is locked
                      (hiddenWhenLocked). A departed suspect must not leave
                      an invisible but clickable conversation. Keep normal
                      locked doors visible so their refusal is discoverable.
- speed               walk speed in world-%/second (defaults to the room's
                      NPC pace — deliberately slower than the player, so a
                      fast player never turns every barman into a sprinter).
- idleAnim            SET MOTION at rest instead of plain "idle": the
                      barman's "polish", the mechanic's "hammer".

THE PATROL
----------
patrol[] is a list of waypoints: walk to each in turn, PAUSE `pause`
seconds there, play that stop's SET MOTION (`anim`) and facing, then
repeat per patrolMode:
  loop      back to waypoint 1 (default)
  pingpong  walk the route backwards to the start, and out again
  once      walk it once and HOLD the last stop's motion forever
Each waypoint may carry:
  x, y      the stop
  pause?    seconds idling before moving on (default 2)
  anim?     the named sheet animation played while paused there
  facing?   direction during the pause (default: direction of arrival)
  say?      an ambient line floated OVER the actor's head — SCUMM
            background chatter; a speech line, not a dialogue beat
  scale?    size multiplier ON ARRIVAL, interpolated along the walk leg —
            a route into the distance shrinks the actor, a route toward
            the camera grows them
TIP: a SINGLE waypoint at the spawn position makes a STANDING actor that
cycles a motion (and ambient line) on a timer — SCUMM set-dressing with
zero walking. That is how Dektora never stops dancing at Early Q's.

INTERACTABLE NPCs (talkHotspotId)
---------------------------------
Point talkHotspotId at a start-conversation marker IN THIS SCENE and the
actor BECOMES that marker: the static diamond hides, the walker chases the
actor's LIVE position (re-routing while it patrols), and the conversation
fires on catch. Pressing N in play walks to the nearest linked character.
Rules that keep this honest:
- The linked marker MUST be a start-conversation action — the validator
  warns if it points at a giveItem/showText/go marker.
- The marker keeps its one-shot / repeatable rules. A first meeting that
  grants a reward is ONE-SHOT (the actor stays on stage afterwards as set
  dressing — the diamond is gone, the person is not). Gossip with no
  reward can be repeatable.
- Ambient (unlinked) NPCs stay pure set dressing, exactly as before.

ONE ROOM, MANY CONVERSATIONS (TOPIC DIALOGUE)
---------------------------------------------
A scene's dialogueBeats are a SHARED queue: every start-conversation
marker in the room plays the WHOLE scene's beats. That is right for a
single-speaker room (Howie's noodle bar). When a room has SEVERAL
speakers (Bryant's office: the chief AND Gaff; Early Q's: Dektora AND
Roy), give each talk marker its own topics[] list instead — the
```

```
   LucasArts pattern:
     walk up -> a HUB of askable questions -> pick one -> ITS beats play
     -> back to the hub -> ask another, or leave ("Never mind")
   Each topic: id, label (the player-facing question), beats[] (the full
   DialogueBeat machinery — voice, expressions, backgrounds, recalls),
   gating? (hidden until flags/items/trust are ready), repeatable? (default
   false — asked once, then it sits in the hub with a check mark),
   flagSet? (records the ask). The marker's own flagSet/then still apply
   when the conversation OPENS (that is where the badge gets handed over).
   Stories that never set topics are 100% unchanged.

   FIELD CHECKLIST
   ---------------
   [ ] Every NPC's stops sit inside the floor band.
   [ ] Patrolling actors never route through obstacles or door markers.
   [ ] A linked marker is a start-conversation (validator-clean).
   [ ] Multi-speaker rooms use topics; single-speaker rooms may use beats.
   [ ] One-shot reward conversations keep the actor on stage after they end.


================================================================================
   SECTION 20  —  WALK-BEHIND PROPS & FOREGROUND DEPTH (STAGE ROOMS)
================================================================================

   A stage scene's foreground extras (scene.cinematic.foregroundExtras[]) are
   separate cutout images composited into the room. Most are dressing; one
   flag turns any of them into a WALK-BEHIND OCCLUDER — the LucasArts trick
   where the counter is a real object: while the player's feet sit ABOVE the
   object's baseline line, the art is drawn OVER the player, so they really
   are behind the counter instead of pasted on top of it. No separate mask
   asset — the cutout you already have IS the mask, alpha and all.

   THE LAYERS
   ----------
   - image / sheet       the cutout, or an ANIMATED sheet (flickering neon, a
                         rain-lashed flag). `sheet` plays its loop on repeat;
                         `image` stays as the static fallback.
   - x, y                where the anchor lands, % of the VIEWPORT (clamped
                         0..100 even in wide rooms). Convention: bottom-
                         centre of the prop.
   - worldX              WIDE STAGES ONLY — pin the layer to a position in
                         the WORLD (0..worldWidth) so it pans with the camera
                         instead of sliding across the room. Overrides x.
   - scale, fit          size the cutout (fit default `contain` hugs the
                         art's ratio). Height = scale x frame height, so the
                         top edge lands at y - scale*100.
   - anchor              the point ON the image pinned at x/y, CSS
                         background-position style; default "50% 100%" = feet
                         on the floor.
   - depth               parallax plane 0 (far) to 1 (near); only drifts when
                         the room has parallax depth on.
   - flip                mirror horizontally (two subjects facing each other).
   - visible, opacity, blendMode, rotation, zoom/zoomDuration
                         compose the shot: ghosts, fog, tilted signs, a slow
                         push-in on the prop.
   - motion / anim       fly the layer along a path (a gull crossing the
                         harbour, a car passing) or loop an idle motion (bob,
                         sway, flicker, flutter) — sampled into CSS
                         keyframes, no per-frame JS.
   - occludes            TRUE = this prop is a walk-behind occluder (off by
                         default; a room can mix occluders and plain
                         dressing).
   - occludeBaseline     where the occluder's base sits, % of PLATE HEIGHT
```

```
                      (default: the extra's own y). The walker is BEHIND
                      it while their feet are above this line — nudge it
                      to move the invisible depth line.
- maskImage / maskFit ALPHA MASK: opacity reveals this layer; transparency
                      hides it. Luminance is ignored. For a cockpit use a
                      WINDOW mask opaque in the glass and transparent over
                      the dashboard — NOT the cabin cutout as the mask.
                      The cabin itself is a separate fixed foreground layer.

SPINNER / TRAIN WINDOW RECIPE (Scene -> Cinematics -> Foreground sprites)
------------------------------------------------------------------------
1. Put the moving scenery on its own foreground-extra layer. Use x=50,
   y=50, anchor="50% 50%" for a full-frame window, with matching maskFit.
2. Set maskImage to the opaque-glass/transparent-cabin mask. Load the fixed
   cabin artwork separately: opaque dashboard, transparent glass. Or reuse
   the full cockpit image with the SAME mask and maskInvert:true. Do NOT
   animate the mask or dashboard when making the view move.
3. Enable scrolling; choose primary axis, speed and optional cross-axis
   speed. Repeating images need genuinely tileable edges. A matching CSS
   loop does not make non-tileable artwork seamless. repeat:false paints
   only one tile; loop:false runs the animation once then holds its end.
4. Start X / Start Y are percent OF THE RENDERED TILE, not the viewport.
   Negative offsets move art up/left. Use them to put the city, not empty
   sky, behind the window. The tile is measured after load and on resize;
   independent X/Y periods move exactly one tile without a loop reset jump.
5. Zero speeds or OS reduced-motion hold a still image with the SAME fit,
   mask and starting offsets. Always test resize and reduced-motion mode.
6. Hide the walker and make the buttons remote. Keep a clear abort/back
   control. In a screen-blended light streak layer use low opacity; a
   normal-blended black road tile can black out an otherwise correct view.


THE WALK-BEHIND RECIPE
----------------------
1. Cut the prop frontal. Key a counter / desk / console plate to
   transparent (SECTION 21 pipeline). Frontal orthographic reads as solid;
   perspective fights the flat room.
2. Place it on the floor. Anchor bottom-centre at x/y; scale until the
   feet sit on the floor. Wide room? Pin with worldX.
3. Draw the depth line. Set occludeBaseline where "behind" should start —
   usually near the MIDDLE of the room's floor band. A floor band of
   68..95 takes a baseline of ~88: far band 68..88 hides behind the prop,
   near band 89..95 walks in front.
4. Sit the actors. An NPC's FAR stop at/above the baseline (it stands
   behind its own counter, head over the edge) and its NEAR stop below
   (it steps out in front). The z-order flips per player position: one
   prop, both depths, no extra art.
5. Test both bands. Walk from far to near and back: the player must
   disappear behind the prop and reappear in front, and markers on the
   near side must stay clickable. Stage controls (markers, sprite
   shadows) render above the prop pass, so a busy foreground never hides
   the interaction layer.

FIELD CHECKLIST
---------------
[ ] Props are anchored at their feet and pinned (worldX) on wide rooms.
[ ] The baseline sits mid-floor-band; NPCs have stops on BOTH sides of it.
[ ] Regenerated prop plates were re-checked for x/y/scale drift
    (generators shift layouts; a prop that floats 3% reads as a hologram).
[ ] The cinematic FOREGROUND OVERLAY (per-shot subject, SECTION 13) is not
    being confused with these room layers.
```

```
================================================================================
   SECTION 21  —  SPRITE SHEETS: ANIMATION & PLATE PREP
================================================================================

   NAVIGATION CHECK: concave floor paths (stairs/L-shaped corridors) route
   around their bends even with NO rectangular furniture. Test all exits from
   every entrySpawns position, especially the return from an upstairs landing.
   A room with one working exit is not proof that its progression exit works.

   STUDIO FORGE: for HD actors choose Preserve pose scale / align feet in
   the sprite-sheet forge. All detected frames use ONE common scale and a
   common foot baseline, so a crouch stays shorter instead of stretching to
   standing height. Legacy per-frame normalization remains the default for
   older assets. Inspect EVERY frame and portrait over light/dark backgrounds:
   no clipped heads/shoes, opaque matte, magenta fringe or changing costume.
   Reduce edge colour spill optionally neutralises key-colour rims on opaque
   edges (e.g. magenta hair outlines) without changing alpha or interior colour.
   Preview and built sheets use the same option; leave it off for legacy art.
   Four illustrated poses are not a complete eight-direction walk cycle.
   Use pixelated:false for smooth photo-style scaling. x/y are feet anchors.

   NEW: CLEAN SILHOUETTES AND MULTI-BANK ATLASES
   ----------------------------------------
   In Upload art -> auto-build sheet:
   - Flat key / retain dark props uses RGB distance for a genuinely flat,
     unshadowed backdrop. It can preserve a dark brown cane that resembles
     shadowed magenta. Leave it OFF for shadowed backdrops (the default chroma
     key handles those). Check the result, not just the detected frame count.
   - Separate sprite islands cuts disconnected silhouettes instead of vertical
     bands. A boot or pot may overlap the NEXT figure's x-range without touching
     it: each frame keeps only its own pixels. Small nearby detached props join
     the nearest body; isolated border specks are dropped. Preview and build
     share the same operation. Touching/occluded people cannot be reconstructed;
     an impossible requested count is refused, not silently invented.
   - Inspect every resulting frame, including soles, hats and narrow props.
     Preserve pose scale keeps crouches short; it cannot repair a wrong cut.

   Below a sheet's clips, open Combine pose banks into this atlas:
   1. Key/cut out each source first. Choose another transparent, evenly gridded
      source using the image browser (built-in or already saved in the library).
   2. Enter that bank's columns/rows, a NEW clip name, FPS and output frame
      height. This source must divide evenly into its grid. Keep transparent
      headroom inside crouch frames; do not tightly crop each pose independently.
   3. Assemble atlas preserves existing clip indices/timing and offsets the new
      clip into a shared, bottom-aligned canvas. It prefers a complete grid,
      bounds the canvas to 8192px, and refuses duplicate clip names. Edit the
      frame lists/loop switch afterwards to turn a bank into several motions.
   4. The result is a new library asset. On a failed save the old sheet stays
      intact; lower output height or raise Media Tools -> Library -> limits.
      Auto-build can instead keep its result inline and now explicitly tells
      you when that did NOT save to the library. Export/back up your work.

   DIRECTIONAL PLAYER CLIPS
   ------------------------
   Base walk / idle keep the old right-facing, mirror-for-left behaviour.
   Optional suffixes: -up, -down, -left, -right, -up-left, -up-right,
   -down-left, -down-right. Add any diagonal WALK clip to enable eight-way
   Stage player headings; four-way sheets keep dominant-axis selection.
   Explicit left art is not mirrored again. Missing left/right angles can
   mirror their opposite; absent diagonals use the vertical angle, then the
   base clip. Missing idle uses a frozen walk frame, not a walk in place.
   Eight heading selections are NOT eight independently photographed angles.
   Named action clips such as duck/draw/holster work with Player pose and
```

```
the existing Stage script pose/walkTo/npcMotion/npcWalkTo tools. loop:false
holds the last frame. Scripts/animation never substitute for earned gates.

CONVERSATION AND STATE REVIEWS
------------------------------
Stage -> Ambient NPCs -> conversation motion (npc.talkAnim) is optional.
It overrides idle/waypoint motion while that NPC's linked conversation is
open, then restores idle. It is a conversation gesture, NOT lip-sync.
A multi-state object's Repetition in this state can inherit, repeat or be
one-shot. Use a one-shot briefing/action, then a flag-gated repeatable
conversation review WITHOUT another reward. showText always repeats.
The LAST matching state wins: put the most specific aftermath state last.
Validator warns if a newly repeatable state can replay rewards.

Everything that moves on a stage — the player's walker, ambient NPCs, the
dialogue-portrait figure — runs on the same SPRITE SHEET: one image holding
a grid of equal frames, plus named animations that pick frame lists and a
frame rate. Photoreal characters look smooth because the frames are
resampled, not because they are pixel art; `pixelated` only flips the
scaling mode (nearest-neighbour for pixel art, smooth for photo work).

THE SHEET
---------
- image                the sheet image (url or `asset:` reference).
- frameWidth, frameHeight
                       the grid cell size in px. The keyer prints the
                       frameWidth when it builds the sheet — that number
                       is what goes in the data.
- animations           named clips, e.g.  idle: frames [0]  and
                       walk: frames [0,1,2,3] at fps 8. Each clip:
                       frames[] (0-based indices), fps? (default 8),
                       loop? (default true; a one-shot clip sets false).
- pixelated            true = crisp pixel art, false = smooth photoreal.

WHO USES A SHEET
----------------
- The player's walker        (story.playerSprite / stage.playerSheet)
- A stage NPC                (npc.sheet — `idle` at rest, `walk` while
                              patrolling; named extras are SET MOTION)
- A cast member's portrait   (character.talkingSheet — the figure in the
                              dialogue box, usually a 2-frame talk pair)
- An animated foreground     (foregroundExtras[].sheet — flicker, flag)
Recipe per character: a 4-FRAME WALK sheet for movement (fps 8) and a
2-FRAME TALK sheet for the box (fps 4) — the same person, two framings.

THE PLATE PIPELINE
------------------
1. SHOOT ON MAGENTA. Generate or paint the character on a flat PURE
   MAGENTA #FF00FF background: a wide plate with N side-by-side poses —
   4 for a walk cycle (contact, down, passing, up), 2 for a talk pair
   (neutral + one gesture). Full body, NO shadow, NO ground line,
   identical costume and face frame to frame.
2. KEY THE PLATE.
      node scripts/make-sprite-sheet.mjs \
        --in plates/hero-walk-plate.png \
        --out public/assets/<case>/hero-walk.png \
        --frames 4 --height 384
   It keys the magenta to alpha, trims, splits uniform frames and prints
   the frameWidth for the data. The Studio's AI image tab runs the same
   magenta-plate to keyed-sheet pipeline, so generated and hand-authored
   plates land in the same place with the same rules. This CLI is the
   legacy uniform-strip path; use Studio's forge/assembler above for the
   new island/flat-key and multi-bank workflows.
```

```
   3. WIRE THE SHEET. Reference the keyed sheet in the data (playerSprite,
      npc.sheet, character.talkingSheet). Walkers get idle + walk; talk
      portraits get idle + talk.
   4. MIND THE FACING. Base walk/idle art is RIGHT-facing and mirrored to
      face left. Stage player directional clips can instead supply explicit
      angles (see above); those must actually face their named direction.
      Verify movement AND rest, plus actions, in Play draft at every angle.

   WHY MAGENTA
   -----------
   A flat bright #FF00FF backdrop is easy to distinguish from many costumes,
   but reflected magenta light can also appear on hair, cloth and dark props.
   Inspect the key and edge spill rather than assuming full tolerance is safe.
   A clean cutout can become a walk-behind mask, portrait or HUD sprite. Repaint
   or use a better source if touching figures cannot be separated cleanly.

   PITFALLS
   --------
   - Frame-to-frame drift (costume, face, scale) reads as a flicker glitch
     at 8 fps — keep ONE costume and ONE face across every frame.
   - Image generators shift layouts between takes: after regenerating any
     plate, re-key it AND re-check every x/y, occludeBaseline and click box
     that was tuned to the old one.
   - A 1-frame "sheet" is a legal cutout: props and walk-behind objects are
     keyed the same way (--frames 1 --height 512).

================================================================================
   APPENDIX A  —  QUICK FIELD REFERENCE
================================================================================

   STORY
     id · title · tagline · coverImage · startSceneId · scenes[] · items[]
     characters[] · deduction? · briefing? · acts? · episodes? · seasonTitle?
     flagNotes?{flag → note} · dialogueBox? · layout? · assets?

   SCENENODE
     id · title · nodeType(standard|credits|cutscene|stage) · background
     partBreak?  (end-of-part route chart label; see Scene tab)
     ambientAudio(none|rain|jazz|drone|tension) · particles(none|rain|smoke|storm)
     musicUrl/sceneAudioUrl · musicMode(inherit|track|none) · narrationAudioUrl
     autoNarrate · noTypewriter · choiceDelaySeconds · credits? · prose
     dialogueBeats[] · hotspots[] · choices[] · onEnterActions[]
     interrogation?  (composure-pressure interrogation; see INTERROGATION)
     isEnding · endingData{tier,headline,recap} · cinematic? · letterbox?
     transition(fade|fade-black|cut) · transitionSpeed?

   CREDITSSETTINGS
     title · subtitle · lines[{role,name}] · font(pulp|poster|elegant|epic|
     modern|type) · cardMotion(blur-rise|letterspace|scale-down|slide|flicker|
     fade|slam) · titleEntrance(fade-up|typewriter|slam) · titleColor?
     style(fade-up|scroll|cards) · cardSeconds? · montage[] · montageSeconds?
     montageTransition(dissolve|cut)? · montageLoop? · stagger? · duration?
     scrollDuration? · accentColor? · gateWeave(off|subtle|worn|projector)?
   NOTE: "type" font is now Newsreader (clean literary serif), not Special Elite.

   CHOICE
     id · text · targetSceneId · attitudeTag(Neutral|Sultry|Aggressive|Clever)
     gating? · consequences? · reply? · hidden? · timeLimitSeconds?
     isRepeatable? · exclusiveGroup? · memoryLog? · quickTime? · qtePrompt?
     qteHeartbeat? · qteShakyCam? · timedOutTargetSceneId? · timedOutConsequences?
     presentItem? · consumePresentedItem?  (the present-item verb; records an
                                            `item:<id>` memory on pick)
     A reply with a non-Neutral attitudeTag in an interrogation scene is a
```

```
  PRESS (chips composure); a Neutral reply is not.

INTERROGATION (SceneNode.interrogation)
  characterId · composure?(1..5, default 3) · crackTrust?(default 0)
  steadySeconds?(3..12, default 7)  pressReactions?(string[])
  brokeBeat{id,characterId,text,expression?} · brokeFlag
  brokeConsequences?  (the big swing on the crack)
  clampedBeat? · clampFlag? · clampConsequences?  (the backfire)

HOTSPOT
  id · label · cursor(look|take|talk|use|go) · bounds{x,y,w,h} · isRepeatable
  gating? · hiddenWhenLocked? · action · customFailMessage? · showItemImage?
  examine? · examineSeconds?  (hold-to-inspect; camera zooms + meter fills)

HOTSPOT ACTION
  type(goToScene|giveItem|setFlag|showText|meterChange|openBoard|useItem|
       patternPuzzle)
  targetSceneId? · itemGranted? · flagSet? · text? · meterChange? · itemConsumed?
  patternPuzzle?{title,instruction,rows[],targetCount,payoffText,allowRetry?,
                 wrongMessage?,rewardConsequences?,wrongSubmissionConsequences?,
                 targetSceneId?,wrongTargetSceneId?}

STAGE (SceneNode.stage)
  spawn{x,y} · speed · reach · keyboard · floor{top,bottom} · depthScale?
  hints · verbCoin · objectVerbMenu · enterCaption · obstacles[]
  npcs[](STAGENPC) · exitEdges{} · entrySpawns[] · worldWidth?
  hotspotTrigger(smart|walkover|aim) · plateAspectRatio? · hidePlayer?

STAGENPC (stage.npcs[])
  id · image?/sheet?(SPRITESHEET) · x · y · scale? · facing?(left|right)
  idleAnim?(set motion at rest) · speed? · gating?
  talkHotspotId?  (a start-conversation marker in this scene — the actor
                   BECOMES it: the diamond hides, the walker chases the
                   actor's live position, N walks to the nearest)
  patrol?[{x,y,pause?,anim?(set motion),facing?,say?(ambient line),
          scale?(arrival size, interpolated)}]
  patrolMode(loop|pingpong|once)

SPRITESHEET
  image · frameWidth · frameHeight
  animations{name:{frames[],fps?,loop?}} · pixelated?
  pipeline: magenta plate -> scripts/make-sprite-sheet.mjs -> keyed sheet
  (base art faces RIGHT; the engine mirrors left-facing sprites;
   walkers idle+walk @8fps, talk portraits idle+talk @4fps)

FOREGROUNDEXTRA (scene.cinematic.foregroundExtras[])
  image/sheet? · fit? · scale? · x?/y?(viewport %) · worldX?(wide rooms)
  anchor?(default "50% 100%" = feet) · flip? · depth?(parallax 0 far→1 near)
  visible? · opacity? · blendMode? · rotation? · zoom?/zoomDuration?
  motion?(path) · anim?(idle loop)
  occludes?(walk-behind) · occludeBaseline?(% plate height, default y)
  maskImage?/maskFit?/maskInvert?(opaque alpha reveals; invert shows outside)
  scroll?{axis,speed,crossSpeed?,repeat?,loop?,offsetX?,offsetY?}
  (walk-behind: walker feet above the baseline = BEHIND the prop;
   see SECTION 20 for the recipe)

DIALOGUETOPIC (a startDialogue action's topics[])
  id · label (the askable question) · beats[](DIALOGUEBEAT)
  gating? (hidden until met) · repeatable? (default false: asked once,
           then shown exhausted in the hub) · flagSet? (records the ask)
  (a marker with topics opens a LucasArts HUB instead of the scene's
   linear beats — one room, many speakers; see SECTION 19)
```

```
   DIALOGUEBEAT
     id · characterId · text · expression? · conditionRule? · background?
     cinematic? · backgroundTransition · backgroundTransitionSpeed?
     audioUrl? · silent? · recalls?{memoryLogTag:line}
     useCharacterAnimation? (default true; false = resting pose, no cycling)

   GATING
     requiresItems[] · requiresFlags[] · excludesItems[] · excludesFlags[]
     requiresAll(default true) · minMeters{} · maxMeters{} · minTrust{}
     requiresMemories[] (memoryLog tags)

   CONSEQUENCES
     setFlags[] · clearFlags[] · addItems[] · removeItems[] · meterChanges[]
     trustChanges[]

   CHARACTER
     id · name · role · avatarUrl · trustMeter · voice{voiceName?,pitch?,rate?}
     talkingFrames? · talkingFrameDuration? · talkingX? · talkingY?
     talkingScale? · talkingFlip? · talkingRest?

   ITEM
     id · name · icon · iconUrl? · description · flavorText · isClue?
     combinableWith?{targetItemId,resultItemId}

   ENDINGDATA
     tier(victory-gold|victory-silver|gameover) · headline · recap

   CASEBRIEFING
     objective · winConditions[] · loseConditions[] · heatLimit? · heatFailSceneId?

   DEDUCTIONBOARD
     suspectIds[] · connections[{a,b,flagSet,revelation}]


================================================================================
   APPENDIX B  —  THE SINGLE "SOFTLOCK" CHECKLIST
================================================================================

   A softlock is a case state the player cannot leave. Before you publish, walk
   these:

   [ ] The start scene exists and is set.
   [ ] Every scene is reachable from the start.
   [ ] Every non-ending scene has at least one open (ungated) exit,
       OR the only way out is gated on something the player can obtain
       earlier in the same run.
   [ ] Every item a gate requires is actually granted somewhere
       (hotspot giveItem, or a choice's addItems, or a recipe result).
   [ ] Every flag a gate requires is actually set somewhere
       (a choice, a hotspot, or a corkboard connection).
   [ ] There is at least one victory ending, and it is reachable.
   [ ] There is at least one game-over ending.
   [ ] The final accusation is gated only on deductions/rewards that can
       really be earned.
   [ ] No repeatable choice/hotspot grants a reward (no farming).
   [ ] Walk across every exit/pickup without activating it in aim mode.
   [ ] Keyboard/empty-ground commands cancel pending interactions.
   [ ] All entry spawns have a usable exit and an approach to required clues.
   [ ] Revisit after NPC gates change; absent actors have no ghost markers.
   [ ] Masked scenery moves, cabin stays fixed; resize/reduced-motion work.
   [ ] Download/export, reimport and publish; play with original art blocked.

   Run the Validator and Repair, then Play draft. If a path still strands the
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player, the Test/Playtester tab will show exactly which gate never opens.

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END OF MANUAL — SCARLET REEL STUDIO MODE

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